Population Genomic SNP Sampling and Geometric Morphometrics of the Black Spotted Topminnow in the Ozark Highlands

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ABSTRACT

POPULATION GENETICS AND GEOMETRIC MORPHOMETRICS OF THE BLACK SPOTTED TOPMINNOW IN THE OZARK HIGHLANDS

by

ERIC WESTHAFER

Chairperson: Professor David D. Duvernell

The central focus in evolutionary biology is understanding how environments influence population genetic diversity and divergence through both adaptive and non-adaptive processes. Using a population genomic approach to examine population structure and morphological divergence of *Fundulus olivaceus*, the black spotted topminnow, allowed us to survey how populations vary in river environments. Fish throughout the river continuum occupy different habitats that vary in several environmental aspects such as water velocity, predators, prey, and spatial heterogeneity. Body shape and size are great features to focus on for identifying adaptations to local environmental pressures. Shape and size have effects on maneuverability and energy conservation and, due to this, are targets for selection to act upon. Populations in differing habitats have shown variation in body shape and size, although not all species show differences between habitat types, and some species do not show consistent changes in morphological characters.

A total of 283 fish were sampled from headwater and large river environments in the Gasconade (150 individuals) and Meramec (133 individuals) river systems during the

summer months of 2013. Individuals were genotyped, and 5744 single nucleotide polymorphism (SNP) loci were used to analyze population structure. In general, the study supported divergence between the two drainages, but some evidence of mixed ancestry for individuals sampled in the large river environments suggesting incomplete lineage sorting or secondary contact. Substantially more genetic structure was observed in the Gasconade River samples when compared to the Meramec River samples. Headwater populations exhibited clear divergence from larger river populations. Headwater divergence could be attributed to smaller effective population sizes and greater susceptibility to genetic drift.

Adult individuals were subjected to geometric morphometric analysis. Of the 283 individuals sampled, 175 individuals were large enough for shape and size analysis. Centroid sizes (measure of overall size) confirmed that headwater fish from both river systems were larger than their downstream large river counterparts. As expected, most significant shape variation was found in the sexual dimorphism between males and females, where males have larger dorsal and anal fins than females. With sexual dimorphic characteristics factored out, Gasconade individuals exhibited deeper bodies than that of Meramec individuals, but Meramec fish exhibited longer caudal peduncles. We were unable to successfully identify outlier loci that could potentially influence observed changes in body morphology.

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TABLE OF CONTENTS

ABSTRACT ii

ACKNOWLEDGEMENTS iv

LIST OF FIGURES vii

LIST OF TABLES viii

Chapter

I. INTRODUCTION 1

Research Objectives 4

II. REVIEW OF LITERATURE 6

Genome Wide Population Studies 6
Fundulus notatus Species Complex 8
Environmental Variation in River Drainage Systems 10
Local Adaptation and Species Divergence 10
Morphological Characters 13

III. MATERIALS AND METHODS 15

Sample Collection 15
DNA Extraction and Sequencing 18
HapMap and SNP Analysis 18
Data Analysis 19
Analysis of Molecular Variance 19
Principal Component Analysis 19
STRUCTURE 20
Phylogenetic Tree 19
Outlier Locus Detection 20
Geometric Morphometrics 21
Size and Shape Assessment of Full Body 21
Shape Assessment of Head, Body, and Caudal Peduncle Separately 23

IV. RESULTS 24

Population Variation Based Upon SNP Data 24

Genetic Variation 24
Analysis of Molecular Variance 27
Principal Components Analysis 28
STRUCTURE 29
Outlier Loci 32
Geometric Morphometrics 34
Analysis 35
Body Size 35
Body Shape 37
Head, Body, and Caudal Peduncle Shape Analysis 41

V. DISCUSSION 50

Variation Within and Among Drainages 50
Outlier Loci 52
Morphological Variation in Size and Shape 53

REFERENCES 57

LIST OF FIGURES

Figure Page

1. Distribution Map of *F. olivaceus* 9

2. Map of Sampled Sites.....17

3. Photograph of *F. olivaceus* with landmarks 22

4. Principal Components Analysis of SNP data 28

5. STRUCTURE Plots.....31

6. Neighbor Joining Phylogenetic Tree 32

7. Plot of F_{ST} Vs Heterozygosity 33

8. Polymorphic Outlier Loci Detected in Bayescan 34

9. Bi-variate Plot of Males and Females from the Gasconade and Meramec 36

10. Principal Components Analysis of Relative Warp Scores from Full Body.....39

11. Thin Plate Spline Representation of Full Body.....40

12. Shape Deformation Representation of Full Body 41

13. Thin Plate Spline Representation of Head Region.....43

14. Shape Deformation Representation of Head Region 44

15. Thin Plate Spline Representation of Body Region.....46

16. Shape Deformation Representation of Body Region 46

17. Thin Plate Spline Representation of Caudal Region 48

18. Shape Deformation Representation of Caudal Region.....49

LIST OF TABLES

Table Page

1. Specific Locations of the Collection Sites 16

2. Sampled Sites with Cumulative Drainage Areas 17

3. Number of Polymorphic Loci & Percent Polymorphism 25

4. Average Heterozygosity Observed in Each Population 25

5. F_{ST} values Between each Site in Gasconade and Meramec Drainages 26

6. Analysis of Molecular Variance of SNP Data 27

7. Analysis of Variance on Centroid Size 36

8. Analysis of Variance on Full Body 38

9. Analysis of Variance on Head Region 42

10. Analysis of Variance on Body Region 45

11. Analysis of Variance on Caudal Peduncle Region 47

CHAPTER I

INTRODUCTION

The central focus in evolutionary biology is understanding how environments influence population genetic diversity and divergence through both adaptive and non-adaptive processes. Population genomics is an emerging discipline combining concepts and technologies of genomics with the goal of understanding how environmental processes affect evolution (Black et al., 2001). In recent years high-throughput sequencing techniques have revolutionized the field of population genetics allowing researchers to observe snapshots of the entire genome (Hohenlohe et al., 2010). Population genomics can broadly be used to study numerous loci or genomic regions to further understand evolutionary processes such as migration, gene flow, and genetic drift. Evolutionarily, all neutral loci across the genome will be affected by environmental factors in a similar manner, whereas loci under selection will behave differently and reveal patterns of variation (Luikart et al., 2003). Examining the underlying genetic basis of local adaptation in populations will clarify our understanding of natural selection, how species adapt to environmental changes, and provide insight into speciation and divergent processes (Deagle et al., 2011).

Using a population genomic approach, it is possible to observe genome-wide patterns of genetic variation across populations that act along continuous riverine habitat transitions. Watershed systems are ideal for studying adaptation along gradients due to the dendritic nature of headwater drainages with fast flowing waters and little predation, moving to mid-stream sites where water flow rates are reduced with increased predation. Large river sites have slow moving water and the biotic community includes large piscivorous predators

(Vannote et al., 1980). Previous allozyme studies have shown high levels of genetic differentiation across populations within river drainages and between drainages (Hänfling & Brandl, 1998). There are two possible processes that can occur in headwaters of drainages: 1) headwater populations exhibit smaller effective population sizes that are geographically isolated, and more susceptible to flow variability due to seasonal effects such as flooding and drought conditions (Gomi et al., 2002) and 2) the unique environments of headwaters imposes distinct selection pressures (faster flow regime, little predation) on populations compared to downstream populations. Using single nucleotide polymorphisms (SNPs), researchers can interpret population genetic structure and detect regions of the genome under divergent selective pressure between local populations to identify outlier loci, where allele frequencies differ significantly from non-outlier background levels (Foll & Gaggiotti, 2008). By identifying outlier loci, researchers may be able to elucidate corresponding phenotypic relationships between populations.

Shape and size differences of individuals in populations can be attributed to a combination of genetic divergence and phenotypic plasticity of which natural selection acts upon (Wimberger, 1992). Fish body shape is directly linked to fitness factors such as swimming performance, prey capture, energy consumption, and predator avoidance (Langerhans & Reznick, 2009). Flow regimes are believed to be a strong selection pressure where body shapes reflect local habitat and the general prediction is that individuals tend to be more “streamlined” in river/stream habitats (Langerhans, 2008). In river systems, clear gradients moving from headwater habitats to large river habitats have been observed, but morphological characters do not always transition in similar ways (Brinsmead & Fox 2002). Hass et al. (2010) found consistent changes in body shape, contrary to conventional beliefs,

where the blacktail shiner (*Cyprinella venusta*) populations in reservoirs exhibited deeper bodies and smaller heads than river fish, indicating a more streamlined body structure in lakes than in river. Topminnows have been the subject of several morphological studies to better understand plasticity and variability based upon habitat (Newman, 1907; Nabrit, 1929; Able & Felley, Thomerson, 1966; 1986; Schaefer et al., 2011). The *Fundulus* genus of topminnows are a good system for studying corresponding genetic and morphological characters to allow insight into environmental factors that affect genetic variation and morphology. *Fundulus* topminnows are abundant, broadly distributed, and occupy a large range of freshwater and marine habitats. Welsh et al. (2013) explored body size and shape differences between lake and stream populations of *Fundulus notatus*, the blackstriped topminnow, and demonstrated that stream fish had larger dorsal fins for maneuvering in the water column and a larger overall size than lake fish. Schaefer et al. (2011) sampled two sister species of sympatric topminnows from headwaters and large river environments to study if the two species converged upon similar morphological characteristics in similar habitats. In contrast to Hass' findings, Schaefer et al. concluded both species converged on a deep body form in headwater habitats, but only *F. notatus* exhibited a shallow body form in large river habitats.

The black spotted topminnow, *Fundulus olivaceus*, exhibits a large range of distribution from southeastern Texas to central Missouri. In the Ozark region of south-central Missouri, *F. olivaceus* populations are distributed throughout the Meramec and Gasconade river systems from intermittent headwater streams to the margins of large rivers, with similar geography and habitats making them suitable systems for study (Thomerson,

1966). Phylogeographic studies suggest a Pleistocene range expansion for *F. olivaceus* which arrived at current distributions through dispersal (Wiley & Mayden, 1985).

Research Objectives

In this research study, I wish to quantify the impact of neutral (gene flow and genetic drift) and non-neutral (selection) processes in partitioning variation among populations and how habitat can influence body shape and size. The major objectives and hypotheses were:

Based on phylogeographic structure within and between drainages, headwater populations should exhibit marked divergence from adjacent large river populations. Headwater divergence may be driven by both neutral and non-neutral processes such that:

1) Neutral evolutionary processes will occur independently in each headwater site, resulting in divergence of each headwater site from all other sampled sites.
2) Non-neutral selection pressures specific to headwater environments should impact all headwater sites in the same manner resulting in parallel divergence of all headwater populations from adjacent large river populations, even across isolated drainages.

To further expand upon the genetics of populations throughout the drainages, a secondary goal was to determine how environmental factors and location in the river continuum affect morphological variation in the overall body, as well as specific body regions (head, body, caudal peduncle). The major hypotheses were:

1) Headwater individuals will exhibit a larger overall size than downstream large river individuals because of a lower predation pressure.

2) Headwater individuals should exhibit a deeper, more streamlined body for sustained swimming and large river individuals will exhibit a more shallow body form for better maneuverability and burst swimming.

CHAPTER II

REVIEW OF LITERATURE

Genome Wide Population Studies

Next-generation sequencing (NGS) methods have made studying large scale DNA sequencing of organisms both inexpensive and reliable (Nielsen et al., 2011). The ability to sample entire genomes of organisms has grown greatly over the last few decades, but only in recent years advancements in DNA sequencing technologies made genome-wide studies widely available and affordable (Luikart et al., 2003; Beaumont & Balding, 2004). This advancement allows population geneticists the opportunity to transition from a theory-driven field to an empirical, data-driven field where genome-scale data gives new insight concerning population history and the significance of selection pressures (Pool et al., 2010). These advances have made it possible for researchers to better understand non-model organisms and population structures without a fully sequenced and annotated genome. In particular, single nucleotide polymorphisms (SNPs), serve as heritable genetic markers that can be measured in single or multiple populations (Davey et al., 2011). The first widely used next-generation sequencing (NGS) technique, restriction-site-associated DNA sequencing (RAD-seq), relies on the use of restriction enzyme-recognition sites randomly distributed throughout the genome (Baird et al., 2008). A recent modified method of RAD-seq that reduces handling and processing, known as genotype by sequencing (GBS), has made SNP sampling more available for a diversity of organisms (Elshire et al,. 2011). Low density genome sampling techniques, such as RAD-seq and GBS, are being employed to construct haplotype maps for use in population wide genomic studies (Elshire et al., 2011). Since GBS and related sequencing strategies do not require *a priori* genomic information, high density

quantitative trait loci (QTL) mapping and genome-wide association studies are being employed to study phenotypic variation in many non-model species (Parchman et al., 2013). SNP sampling has been instrumental in allowing geneticists to inexpensively obtain a low resolution genome wide map using SNPs to identify population structure, estimate F_{ST}, examine genetic variation within populations, and examine regions of the genome under divergent selective pressures. Divergent loci across pairs of populations could provide strong evidence of selection pressures that vary by habitat (Williams & Oleksiak, 2011). Once candidate outlier regions are identified, more expensive higher resolution sequencing can be performed to better examine divergent selection signatures as well as identify quantitative trait loci in the genome (Narum et al., 2013).

One of the main purposes in studying genetic variation among populations is to identify certain genes that are important for fitness and adaptation (Luikart et al., 2003). Consequently, it is important to identify non-neutral loci under selection showing departure from neutral background levels of neutral loci. Neutral loci are not subject to selective pressures from the environment, but are subject to random processes such as genetic drift. Limitations to these programs are known for having a propensity in identifying false positives, making it important for researchers to closely scrutinize loci exhibiting divergence (Villemereuil et al., 2014). Once outlier loci are identified, researchers can thoroughly evaluate loci and associated genomic regions and even compare them with morphological characteristics.

One such species, *Fundulus heteroclitus*, has shown great promise as a non-model organism in genome wide studies. Continuous populations have been described from as far north as Canada and as far south as Florida with strong geographic clines (Powers et al.,

1986). Populations in the *Fundulus* genus have rapidly adapted to a range of local habitats which consist of polluted, hypoxic, marine, freshwater, and temperate water conditions making them exceptional specimens to study evolutionary trends (Wirgin & Waldman, 2004; Diaz & Rozenberg, 1995; Diaz, 2001; Smith & Abel, 2003). Genome wide studies have allowed researchers to identify regions of the genome where strong selective pressures have shaped ecological responses to anthropogenic and natural environmental changes allowing populations to thrive in imperfect conditions (Deagle et al., 2011).

Fundulus notatus Species Complex

The *Fundulus notatus* species complex consists of three species of topminnows: The blackstriped topminnow *F. notatus*, the black spotted topminnow *F. olivaceus*, and the broadstripe topminnow *F. euryzonus* (Thomerson, 1966; Black and Howell, 1978). Both *F. olivaceus* and *F. notatus* exhibit a large range of distribution from southeastern Texas through the Mississippi River watershed and *F. notatus* populations extend as far north as Ohio. (Thomerson, 1966) (Figure 1). In the Ozark Highlands, *F. olivaceus* solely populates river drainages from intermittent headwaters to large river confluences. The Gasconade and Meramec sites were selected to identify and quantify genomic regions subject to convergent selection pressures among headwater and large river populations and to compare morphological characteristics that may have evolved independently in each drainage based on similar local habitats.

Fundulus olivaceus is a species of fish belonging to the order Cyprinodontiformes and the family Fundulidae. They range in size from approximately 30mm to 85mm (Braasch & Smith, 1965) and are characterized by a broad black lateral-line stripe with an olive-tan color located dorsally of the stripe and a lighter coloration on the ventral surface. The species

is identifiable by varying densities of black spots above the lateral-line on the dorsal surface, dorsal fin, and caudal fin. If found to be in close niche competition with *F. notatus*, *F. olivaceus* generally occupy headwater habitats with clear water, fast stream flow, little vegetation, and gravel substrate (Schaefer et al., 2009). Major food items of *F. olivaceus* include terrestrial arthropods, aquatic insects, and filamentous algae (Thomerson & Wooldridge, 1970). The diet of *F. olivaceus* reinforces corresponding morphological characters associated with surface-feeding fish, including a supra-terminal mouth, dorsally oriented eyes, and a sagittiform body type.

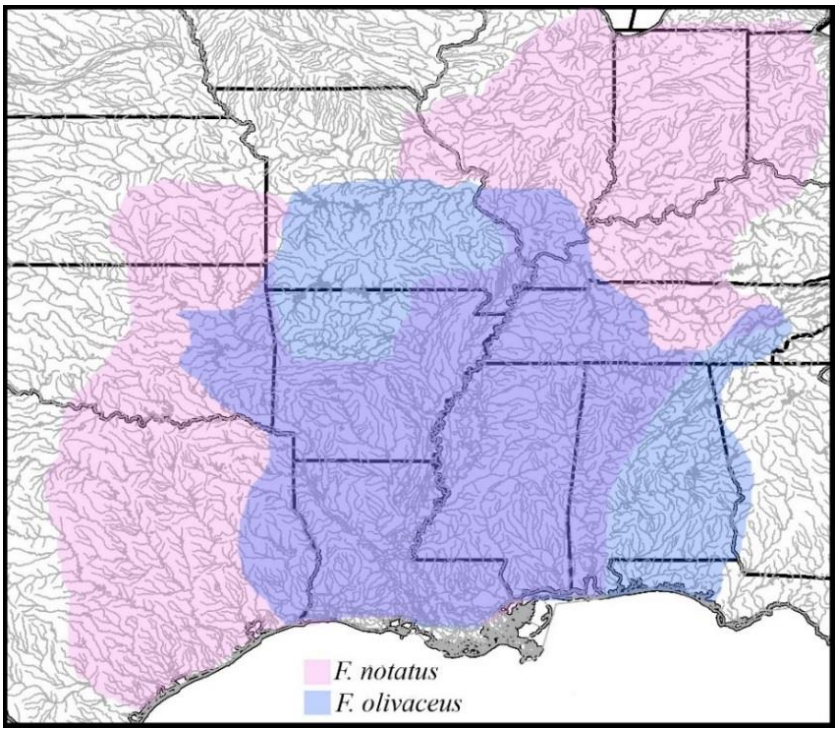


Figure 1. Distribution map *F. olivaceus* (blue) from eastern Texas to central Missouri

Environmental Variation in River Drainage Systems

The River Continuum Concept states that biological characteristics and processes of a river change in a predictable manner while moving from headwaters to downstream habitats (Vannote et al., 1980). The dendritic nature of a river system is broken down into orders (1-7) to characterize changes moving from headwaters to large river habitats. Lower order streams (1-3) characterize headwater habitats, mid-order streams (4-6) are intermediate waters between headwater and large river systems, and the lower reaches (>6) indicate large river environments and confluences of larger rivers (Vannote et al., 1980).

Headwater habitats are often more shallow with cool, swift moving water found in areas of heavier riparian vegetation. Fish often exhibit morphological shapes that enhance their ability to move easily in the water and feed. (Vannote et al., 1980). Energetic inputs are greatest in headwaters and predators are rare because of reduced food and habitat. Mid-reach habitats are often wider and less vegetation is found along the banks allowing for more sunlight to reach substrate within waters. The presence of predators in mid-reach habitats appear due to increased availability of food and larger pools providing cover and habitat (Vannote et al., 1980). Fish found in large river habitats are more adapted to slow moving water and feed on invertebrates, detritus, or other prey fish (Vannote et al., 1980). Predators are abundant in large river environments. Fish distributed throughout all habitats in the river continuum may experience a wide range of environments and selection pressures.

Local Adaptation and Species Divergence

Adaptation via natural selection is the fundamental principle of the theory of evolution. In fact, Wright (1931) assumed the mechanism of evolutionary change through

natural selection would be most effective in species with substantial population subdivision throughout environmental ranges. Detecting genomic signatures of natural selection is key in allowing better understanding of what genomic regions or genes are affected by selective forces (Joost et al., 2007). In theory, forces and constraints found in local demes evolve traits to provide an optimal advantage under local environmental conditions (habitat). An adaptation is generally described as a trait or phenotypic feature that increases the survivability or reproductive success of individuals with the beneficial trait. In most cases, this would require studying adapted populations and less adapted ancestors, but this scenario is usually not feasible (Travisano & Rainey, 2000). On average, it is expected that individuals found within their specific habitat range would exhibit a higher relative fitness than individuals originating outside their habitat range. This relative process is known as local adaptation (Williams 1966). Local adaptation studies have the promise of providing insights into the power of natural selection relating to gene flow, genetic drift, migration and other external forces (Kawecki & Ebert, 2004).

In riverine habitats, genetic structure can be strongly impacted by distance and physical barriers, man-made or natural. Isolation by distance exists when populations become less genetically similar as geographic distance increases (Kimura & Weiss, 1964). With small fish, such as topminnows, migration over large distances is physically costly, although studies have indicated topminnows are capable of traveling on the order of a kilometer or more in just a few days (Alldredge et al., 2011). With fish populations, habitats may not be clearly distinct because of the continuous nature of streams and rivers, but certain conditions within the environmental gradients of the river system describe the defined

habitat. Oftentimes, due to geographical changes, fish are able to invade new waterways, reservoirs, and marine environments where adaptation to local habitats occurs.

Threespine stickleback populations are a prime example of adaptive evolution at the local level. After Pleistocene glaciers retreated, marine sticklebacks colonized and adapted to newly formed freshwater lakes and streams in a relatively short period of time. Recurrent evolution of similar phenotypic characters (shape, loss of skeletal armor, pigmentation, spine size) in similar environments indicated evolution by natural selection (Jones et al., 2012). Deagle et al. (2012) identified that all lake sticklebacks share the same genetic signatures where the fish exhibit phenotypes including increased body length, more gill rakers, longer spines and shallow body depth, and more lateral plates than their stream stickleback counterparts.

Local adaptation is key for the survival of organisms living in harsh and stressful environments. It has been widely studied that salinity tolerance can be extremely taxing or lethal if an organism does not acquire the proper osmoregulatory abilities (Boutilier et al., 1992). The Natterjack toad, *Bufo calamita*, is able to breed in both brackish and freshwater environments where populations have been isolated due to sea currents acting as a strong barrier. Gene flow was substantially affected between populations, and genetic drift has resulted in divergence among local populations. It is believed the ancestral populations lived solely in freshwater environments and subsequent populations have evolved a mechanism to survive in high salinity environments. In common garden experiments, offspring from individuals reared in a high salinity environment demonstrated a significant increase in survivability and successful reproduction, where individuals raised in freshwater

environments and transferred to saline environments displayed increased mortality and low fecundity (Gomez-Mestre & Tejedo, 2003).

Morphological Characters

In terms of morphology and ecology, fish are among the most diverse aquatic biota (Helfman et al., 2009). Body shapes can provide insight into the results of evolutionary adaptations from environmental pressures, most notably food, predation, and hydrodynamic conditions (Matthews, 1998). Phenotypes that deviate from optimal performance are generally suggested to exhibit increased fitness costs (Dodson, 1984). The general prediction is that fish from streams and rivers tend to be more streamlined than individuals from lake or pool habitats (Langerhans, 2008), but there is evidence this is not universal (McGuigan et al., 2003). Differences in size and shape can also be attributed to intrinsic factors, such as sex or age (Parker, 1992). A study by Welsh et al. (2013) demonstrated *Fundulus notatus* of the same age were larger in stream habitats than lake habitats. It is believed size differences occur because lake habitats have high predation of larger fish/adults that select for a smaller size at maturation (Charlesworth, 1980). *Fundulus notatus* and *Fundulus olivaceus* are sister species that are capable of occupying the full spectrum of a drainage, but when found in sympatry, *F. notatus* generally occupies large river habitats and *F. olivaceus* occupies headwater habitats (Braasch & Smith, 1965; Thomerson, 1966; Howell & Black, 1981). Schaefer et al. (2011) demonstrated that *F. notatus* and *F. olivaceus* converged on similar body shape and size in headwater habitats. In headwater habitats both species displayed a more hydrodynamic shape, with deeper bodies. Downstream *F. notatus* populations had shallower body forms, but *F. olivaceus* did not converge on a shallow body form.

Geometric morphometrics is a method used to quantitatively measure shape and size based on landmark coordinates (Adams et al., 2004). Landmarks are placed at homologous biological markers on the specimen of interest and a variety of statistical methods are applied. The most common method, General Procrustes Analysis (GPA), is used to remove variation in landmark configurations outside of shape. From the GPA, it is possible to statistically measure differences of size (centroid size) and shape (warp scores).

CHAPTER III

MATERIALS AND METHODS

Sample Collection

F. olivaceus samples were collected from the Gasconade and Meramec rivers by dipnet between May and July 2013. Specimens collected were overdosed in MS-222, the right or left pectoral fin was clipped and preserved in alcohol for DNA analysis, then the specimens were immediately preserved in 10% formalin for later morphometric analysis. Efforts were made to collect all sample sites within 2 weeks to minimize the influence of seasonal changes. This is of particular concern for females, which can vary substantially in shape depending on whether or not they are gravid. Four sites in each drainage were sampled including 2 headwater sites and 2 downstream large river sites (Table 1). Gasconade headwater sites, Brushy Creek and Baker Creek, were located near Houston and Mansfield, Missouri, respectively (Figure 2). Downstream sites, the Gasconade mid-stream and Gasconade large river, were located near Jerome and Mt. Sterling, Missouri respectively (Figure 2). Meramec headwater sites, the Big River and Dry Fork, were located near Belgrade and Seaton, Missouri, respectively (Figure 2). Downstream sites, the Meramec mid-stream and Meramec large river, were located near Allenton and Arnold, Missouri, respectively (Figure 2). These sites were selected to give a representation of the full river continuum from headwaters to large river confluences. The sites selected are similar in terms of cumulative drainage area across drainages where Gasconade headwaters represent approximately 4-8.5% of overall drainage area and the Meramec headwaters range from approximately 1-6% of overall drainage area (Table 2). The Gasconade and Meramec mid-

stream sites represent approximately 82% and 71% of overall drainage area, respectively (Table 2).

Table 1. Specific Locations of the Collection Sites (H) represents a site designated as headwater. (D) represents a site designated as downstream large-river.

	Site	Latitude, Longitude	Nearest Town (MO)	Number of Individuals	Collection Dates
1	Big River (Meramec) (H)	N37° 45.23' W90° 53.077'	Belgrade	35	5/14/2013
2	Dry Fork (Meramec) (H)	N37° 51.196' W91° 40.041'	Seaton	20	5/14/2013
3	Meramec Mid-Stream (D)	N38° 28.88' W90° 37.826'	Allenton	40	6/11/2013
4	Meramec Large River (D)	N38° 26.252' W90° 20.781'	Arnold	38	7/20/2013
5	Brushy Creek (Gasconade) (H)	N37° 19.980' W91° 57.454'	Houston	45	5/14/2013
6	Baker Creek (Gasconade) (H)	N37° 10.121' W92° 36.866'	Mansfield	40	5/15/2013
7	Gasconade Mid-Stream (D)	N37° 56.296' W91° 58.779'	Jerome	30	5/27/2013
8	Gasconade Large River (D)	N38° 33.81' W91° 35.890'	Mt. Sterling	35	5/27/2013

Table 2. Sampled Sites with Cumulative Drainage Areas

Sampling Site	Cumulative Drainage Area
Brushy Creek	34.67 km^2
Baker Creek	75.41 km^2
Gasconade Mid-Stream	7,251.05 km^2
Gasconade Large River	8,816.66 km^2
Big River	48.57 km^2
Dry Fork	615.47 km^2
Meramec Mid-Stream	7,260.14 km^2
Meramec Large River	10,235.62 km^2

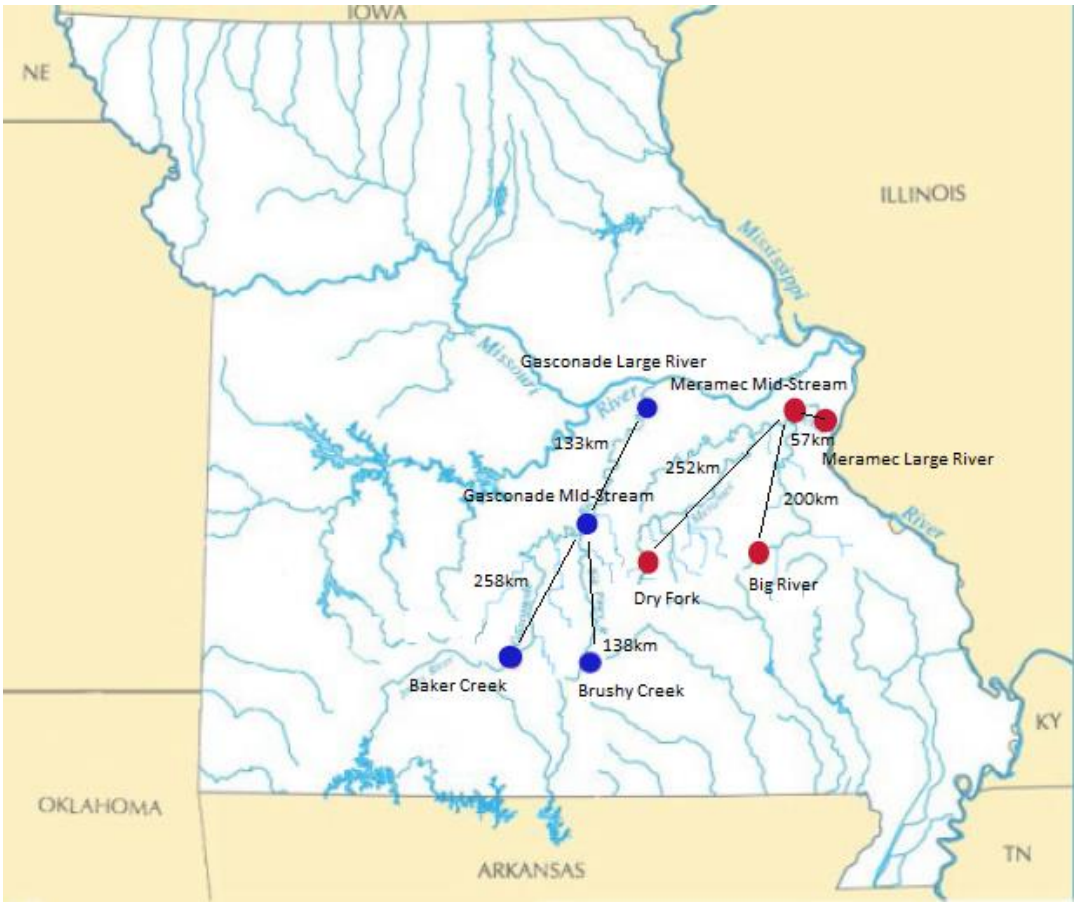


Figure 2. Map of Sampled Sites. Gasconade River sites in Blue and Meramec River sites in Red. Geographic river distances listed between each site

DNA Extraction and Sequencing

DNA extraction of fin clips was performed using the DNeasy® miniprep kit following manufacturer's specifications (QIAGEN). DNA was checked for quality via gel electrophoresis on a 1% gel, and electrophoresed at 95 volts. DNA was quantified and diluted to 50ng/uL, then submitted for Genotype By Sequencing (GBS) following recommended protocols from the Institute for Genomic Diversity at Cornell University (http://www.biotech.cornell.edu/brc/genomic-diversity-facility). Genomic DNA was paired with specific identifier adapter pairs to identify individuals and digested using the restriction-enzyme ecoT22I. Barcode adapters were ligated to restriction enzyme digest fragments then pooled together and amplified for library creation. Pooled libraries were then sequenced using the Illumina Genome Analyzer II (Illumina, Inc.).

HapMap and SNP Analysis

A haplotype map was created using TASSEL v 3.0 GBS pipeline and contigs were aligned using a reference genome of *Fundulus heteroclitus* (https://my.mdibl.org/display/KW/Home). Average heterozygosity across individuals and loci of each population sample were estimated using POPTREE2 (Takezaki et al., 2010). SNPs with excessive heterozygosity (>55%) were excluded from the dataset. Excessive heterozygosity could occur from sources such as alignment errors, paralog and ortholog overlap and low confidence SNP calling. SNP loci that contained more than 5 percent missing data and individuals with more than 3 percent missing data were removed from the dataset. The proportion of loci that were polymorphic and F_{ST} values were calculated to quantify variation within the populations using POPTREE2 (Takezaki et al., 2010).

Data Analysis

Analysis of Molecular Variance

AMOVA is a method of estimating divergence between populations using molecular data and Boolean vectors. AMOVA calculates the significance of variance and eliminates the normality assumption by removing outliers which is not appropriate for molecular datasets (Excoffier et al., 1992). A haplotype map of the final dataset was converted to the Arlequin file format using PGDSpider v2.0.5.1 (Lischer & Excoffier, 2012). The dataset was analysed using analysis of molecular variance (AMOVA) in Arlequin 3.5 (Excoffier et al., 2005). Variance was compared between the drainages, and among populations within drainages.

Principal Component Analysis

Principal component analysis (PCA) is used on datasets to provide a visual representation of variation seen and to bring out strong patterns. A haplotype map of the final dataset was converted to a STRUCTURE file format using PGDSpider v2.0.5.1 (Lischer & Excoffier, 2012). A PCA was performed on single nucleotide polymorphism (SNP) data to separate populations using the ADGENET package in R-Studio software v 0.98.1091 (Jombart and Ahmed, 2011). R-Studio software is a freebased program for statistical analyses utilizing packages designed by users and shared in the scientific community. Analysis provided percent variation, principal component scores, and graphical representation of data.

STRUCTURE

STRUCTURE is a model-based clustering program that places individuals into a predetermined number (K) of subgroups based on allelic frequencies. Distinctive population ancestry and genetic structure was analyzed using STRUCTURE 2.3 (K=8, 10,000 burn in, 100,000 repetitions) (Pritchard et al., 2000). The appropriate K was selected using STRUCTURE HARVESTER v A.1 (Earl & vonHoldt, 2012) which employs the Evanno method (Evanno et al., 2005). The Evanno method uses a maximum likelihood estimation of each specific K cluster to determine the uppermost value of K that describes a dataset.

Phylogenetic Tree

The STRUCTURE file was further analyzed with POPTREE2 (Takezaki et al., 2010) to construct a neighbor-joining phylogeny of population samples. The 'D_A' neighbor-joining method was utilized, which calculates pairwise differences between all pairs of sequence data and operates under the assumption that genetic distances are due to mutation and genetic drift (Nei & Tateno, 1983). Bootstrap parameters were set to 1,000 to resolve clades and evaluate the statistical support of each node.

Outlier Locus Detection

Outlier locus detection was conducted utilizing BayeScan v 2.1 (Foll and Gaggiotti, 2008). BayeScan estimates probability that a given SNP is under divergent selection by calculating posterior odds. This is the ratio of posterior probabilities of selection and neutral models for each locus. The Gasconade and Meramec rivers were evaluated separately. A stringent 'Prior10' was used to detect outliers, which assumes the neutral model is 10 times

more likely than the selection model. An expected false discovery rate of less than 0.05 was used to reduce false-positive results.

Geometric Morphometrics

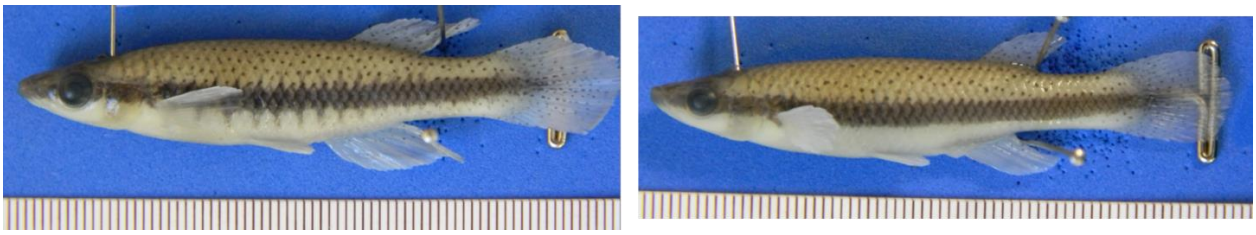
Geometric morphometric (Mitteroecker & Gunz, 2009) techniques were used to study variability in size and shape of *F. olivaceus* from headwater to large rivers across drainages. Only mature individuals were selected to ensure proper sex identification. We evaluated 48 males and 58 females from the Gasconade drainage, and 30 males and 44 females from the Meramec drainage. Sex was determined by examining sexually dimorphic traits including the lateral line, dorsal fin, and anal fin size. The Meramec large river site was collected a month late due to unseasonable flooding of the Mississippi River. Due to the small size of fish collected at the Meramec large river site near Arnold, Mo, these individuals were excluded from the geometric morphometric analysis. Other samples were excluded for excessive curvature of the body during preservation, and females that were gravid with excessive shape changes. The left side of each specimen was photographed using a 12.1 megapixel Nikon COOLPIX L110 digital camera on a background consisting of 1 centimeter by 1 centimeter grids.

Size and Shape Assessment of Full Body

Seventeen landmarks (Figure 3) were digitized on each specimen using tpsDig version 2.17 (Rohlf, 2010). All landmarks used were homologous type I landmarks with the exception of landmark 9, located dorsally over the pelvic fin insertion point of landmark 10 which allows us to capture body depth. Type I landmarks are placed on homologous features of the fish and semi-landmarks are placed to cover regions of the body with no distinctive

features (curves, large areas with no homologous features, etc.). The "Unbend Specimens" tool was used in the tpsUtil version 1.58 (Rohlf, 2013) to remove the "U-bending" artifact that occurs in the preservation process of the fish (Wund et al., 2008).

A.



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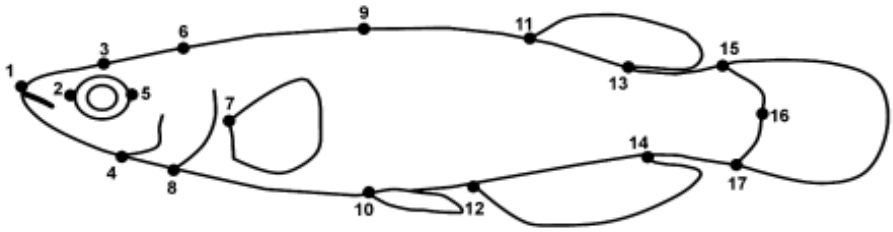


Figure 3. A. Photograph of *F. olivaceus* with landmarks. Fish are pinned to show articulation points of dorsal and anal fins at landmarks 13-14, and a pin to identify landmark 6. Left photograph is a male topminnow with large dorsal and anal fins with jagged lateral line. Right photo is a female topminnow with small dorsal and anal fins and straight lateral line. **B.** Landmarks digitized on all fish for geometric morphometric analysis.

A Generalized Procrustes analysis (GPA) was performed in tpsRelw version 1.53 to remove variation in landmark configurations and produce a set of shape coordinates for each specimen (Mitteroecker & Gunz, 2009). This produced a measurement of size (centroid size) and shape (relative warp scores) that are independent of one another. Thin plate splines in tpsRelw allowed visualization of shape changes that occurred along each principal

component axis. Data obtained from tpsRelw were loaded into Past v 2.17c (Hammer, 2010) to create shape deformation diagrams of each principal component.

Body size was measured via centroid size. A principal components analysis using relative warp scores was used as a measure of body shape. We considered the first four principal component axes because individually they accounted for at least 5% of the variation and together accounted for approximately 65% of the variation.

Shape Assessment of Head, Body, and Caudal Peduncle Separately

All landmarks digitized from above were used to assess measurements of head, body, and caudal peduncle. Using tpsUtil, landmarks not associated with each region of the body were removed accordingly. For the head region, landmarks 1 through 5 were used to analyze size (centroid size) and shape (PC scores). In the body region, landmarks 6, 9, and 11 were located on the dorsal side, and 8, 10, and 12 were located ventrally. To determine caudal peduncle size and shape, landmarks 13 through 17 were used. A GPA removed all variation in landmark configuration to produce relative warp scores and centroid sizes as repeated from above.

The head region was analyzed using the first two principal components that individually accounted for at least 20% of the variation and together accounted for 66% of the variation. The body region was analyzed using the first four principal components that individually accounted for at least 20% of the variation and together accounted for 66% of the variation. The caudal peduncle region was analyzed using the first four principal components that individually accounted for at least 20% of the variation and together accounted for 64% of the variation.

CHAPTER IV

RESULTS

Population Variation Based Upon SNP Data

Black spotted topminnow DNA was genotyped at 18896 biallelic SNP loci from 281 individuals using Illumina's Genome Analyzer II (Illumina, Inc.). After filtering loci containing excess heterozygosity, loci with missing data, and individuals with missing data, the final dataset containing 5744 SNP loci from 248 individuals was analyzed using various methods to differentiate population structure.

Genetic Variation

For each of the populations, a series of analyses allowed us to observe variation among populations in both the Gasconade and Meramec rivers. By observing percent polymorphic loci, we can elucidate which sites in drainages have been more susceptible to genetic drift and fixation of certain alleles. In the Gasconade, the headwater sites, Brushy Creek and Baker Creek, have moderately fewer polymorphic loci than all other sites (Table 3). The headwater sites in the Meramec do not follow this same trend. Average heterozygosity was also calculated for each site, and the Meramec sites along with the Gasconade Large River site display slightly larger values (Table 4). Global F_{ST} values provided information on approximately how much genetic differentiation is occurring between each site, and between drainage sites. F_{ST} values were greatest between the Gasconade and Meramec drainages, and within drainages, headwater sites displayed the greatest divergence (Table 5).

Table 3. Number of Polymorphic Loci & Percent Polymorphism

Collection Site	# Polymorphic Loci	% Polymorphic Loci
Brushy Creek	3043	52.97
Baker Creek	2932	51.04
Gasconade Mid-Stream	3588	62.46
Gasconade Large River	4110	71.55
Big River	3906	68.00
Dry Fork	3538	61.59
Meramec Mid-Stream	3923	68.29
Meramec Large River	4280	74.51

Table 4. Average heterozygosity observed in each population

	Average Heterozygosity	Standard Error
Brushy Creek	0.115	0.002
Baker Creek	0.116	0.002
Gasconade Mid-Stream	0.117	0.002
Gasconade Large River	0.155	0.002
Big River	0.142	0.002
Dry Fork	0.163	0.002
Meramec Mid-Stream	0.161	0.002
Meramec Large River	0.166	0.002

Table 5. F_{ST} values between each site in both the Gasconade and Meramec drainages. F_{ST} values are greatest between drainages, and within drainages between headwaters.

	Brushy Creek	Baker Creek	Gasconade Mid-Stream	Gasconade Large River	Big River	Dry Fork	Meramec Mid-Stream	Meramec Large River
Brushy Creek	-	0.2424	0.0796	0.112	0.3628	0.3349	0.3299	0.3148
Baker Creek	0.2424	-	0.2084	0.1798	0.3511	0.3221	0.3155	0.2977
Gasconade Mid-Stream	0.0796	0.2084	-	0.0574	0.3432	0.3097	0.3062	0.2895
Gasconade Large River	0.112	0.1798	0.0574	-	0.2131	0.1726	0.1669	0.1541
Big River	0.3628	0.3511	0.3432	0.2131	-	0.1096	0.1017	0.0903
Dry Fork	0.3349	0.3221	0.3097	0.1726	0.1096	-	0.0557	0.044
Meramec Mid-Stream	0.3299	0.3155	0.3062	0.1669	0.1017	0.0557	-	0.0144
Meramec Large River	0.3148	0.2977	0.2895	0.1541	0.0903	0.044	0.0144	-

Analysis of Molecular Variance

An analysis of molecular variance (AMOVA) was conducted jointly on both drainages and separately in each drainage. The AMOVA indicated that 20% of variation was partitioned between the Meramec and Gasconade drainages and 9% of variation was partitioned among populations within the drainages. When considered individually, the analyses of the Gasconade River indicated approximately 15% of variation was partitioned among the populations within the drainage. The Meramec displayed approximately half (6.5%) as much variation partitioned among populations as the Gasconade drainage.

Table 6. Analysis of Molecular Variance (AMOVA) of SNP Data. **A.** Variation partitioned between drainages. **B.** Variation among populations within the Gasconade **C.** Variation among populations within Meramec River.

A. Between Drainages

Source of Variation	d.f.	Sum of Squares	Variance Components	% of Variation
Among Drainages	1	31,803.024	114.603	20%
Among Populations within Drainages	6	20,428.526	49.167	9%
Within Populations	488	195,533.564	400.684	71%
Total	495	247,765.115	564.453	

B. Gasconade

Source of Variation	d.f.	Sum of Squares	Variance Components	% of Variation
Among Populations	3	13,737.397	63.53	15
Within Populations	264	94,096.356	356.426	85
Total	267	107,833.754	419.956	

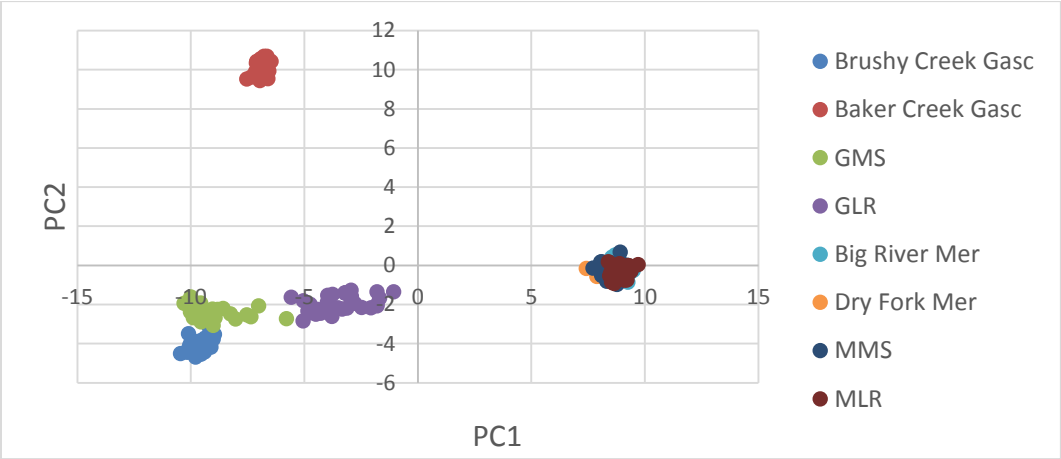
C. Meramec

Source of Variation	d.f.	Sum of Squares	Variance Components	% of Variation
Among Populations	3	6,648.49	31.674	6.5
Within Populations	224	101,011.729	450.945	93.5
Total		107,660.216	482.619	

Principal Components Analysis

PC1 distinctly separates the Meramec and Gasconade River systems. The remaining principal components (PC2-PC5) separated out each of the headwater populations from each drainage. The Gasconade River displays more genetic variability than the Meramec River as seen by clustering of PC scores.

A.



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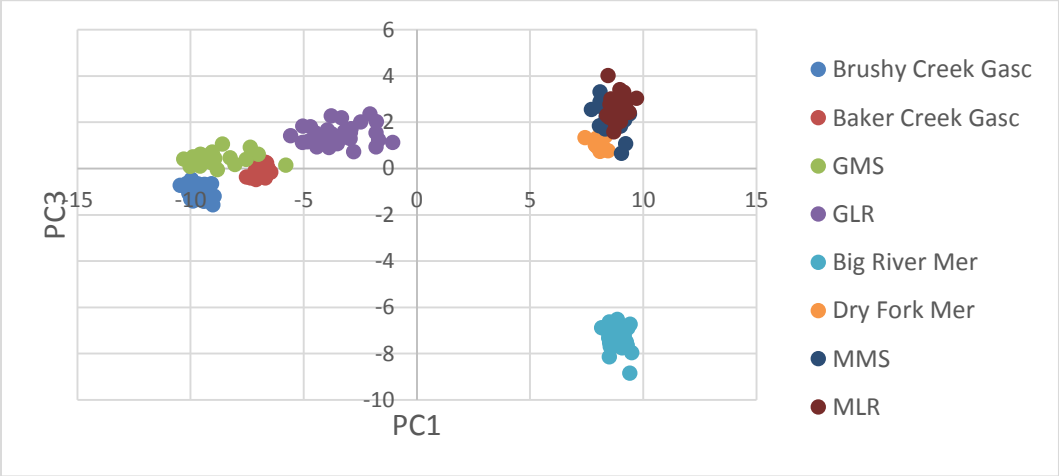

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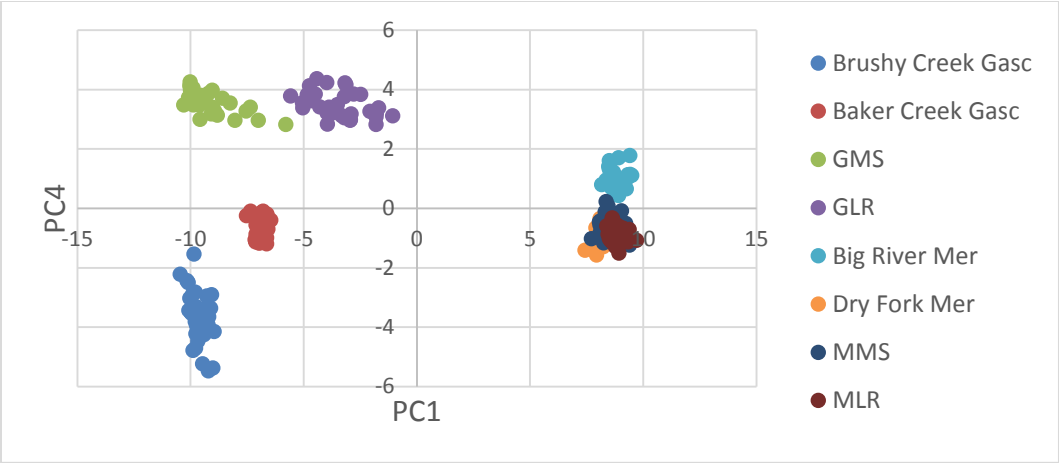


D.

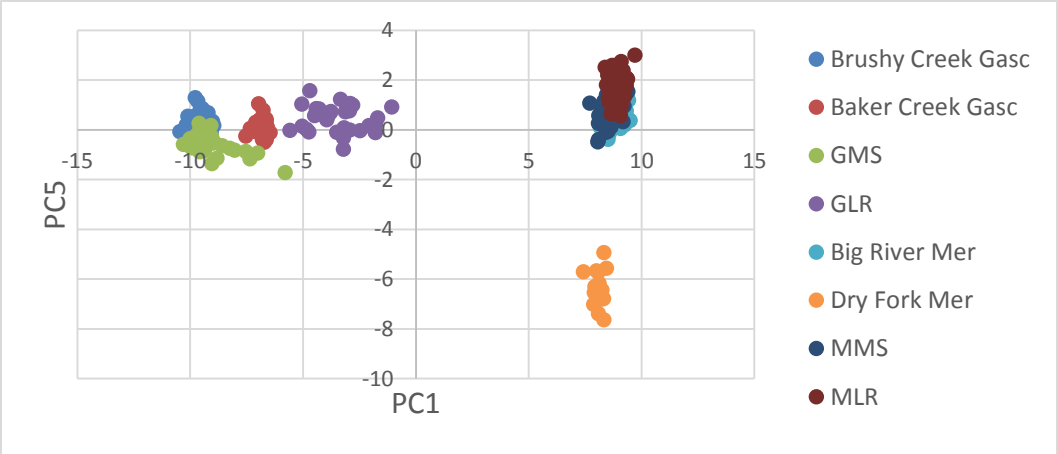


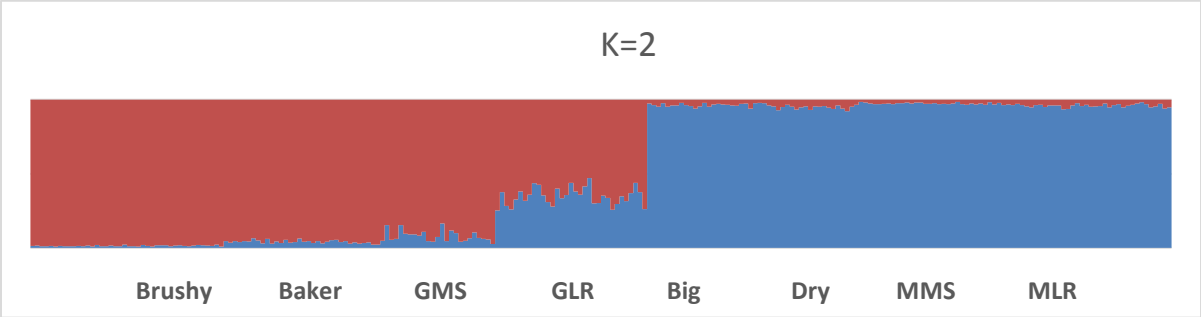
Figure 4. Principal Components Analysis (PCA) of SNP Data **A.** PC1 separates the Meramec and Gasconade drainages. PC2 separates out the Baker Creek headwater site. **B.** PC3 separates the Big River headwater site. **C.** PC4 separates the Brushy Creek headwater site. **D.** PC5 separates the Dry Fork headwater site.

STRUCTURE

SNP data from all eight populations were analyzed in STRUCTURE to form inferred population clusters. At K=2 the clusters distinctively separate individuals from the Meramec and Gasconade drainages with some mixed ancestry in the downstream Gasconade Large

River site (Figure 5a). At K=3 the clustering begins to differentiate the headwater, Baker Creek, site in the Gasconade drainage (Figure 5b). At K=4, clustering builds upon the population structure and the headwater, Big River, site in the Meramec drainage differentiates into its own cluster (Figure 5c). At K=5 the headwater, Brushy Creek, site in the Gasconade drainage differentiates from much of the rest of the Gasconade drainage (Figure 5d). STRUCTURE Harvester using the Evanno method (Evanno, 2005) converged upon an optimal K=2 to represent the data. Given geographical and biologically reasonable clustering observed in higher K-values, a K=5 was selected to provide the best overall representation of genetic structure between the drainages. Overall, more genetic structure is uncovered in the Gasconade drainage than the Meramec drainage and three of four headwater populations showed evidence of genetic divergence.

A.



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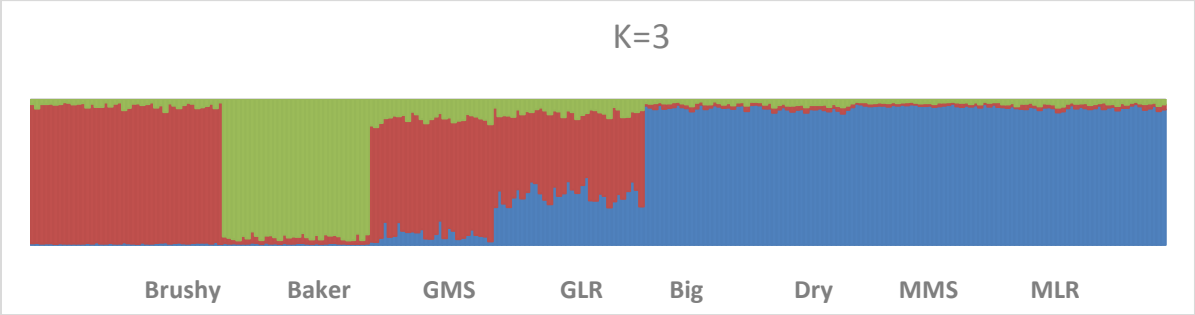

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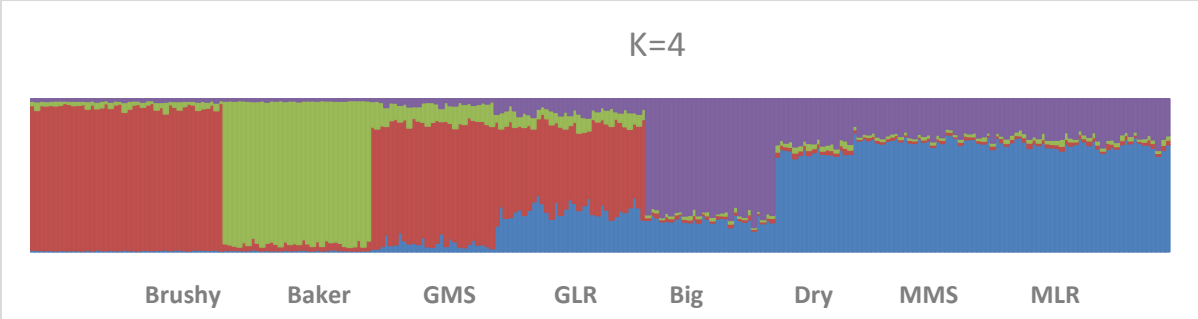

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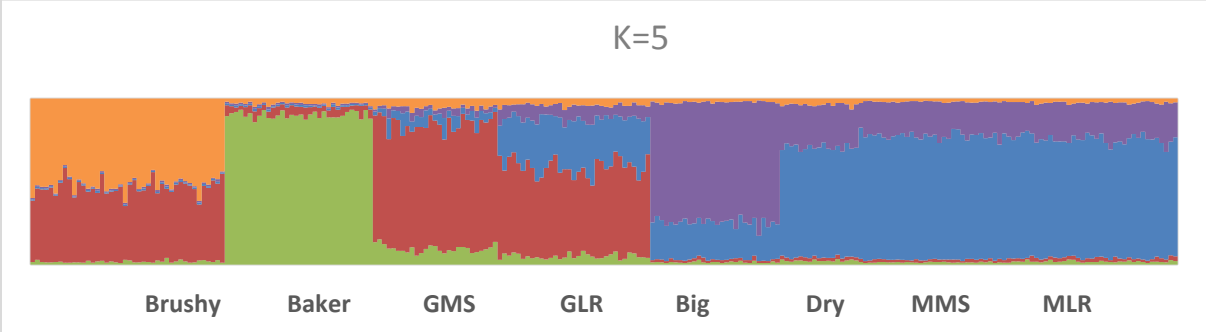


Figure 5. STRUCTURE plots. Analysis of eight sampled populations from Gasconade and Meramec Rivers. **A.** K=2 separated the Meramec and Gasconade drainages. **B.** K=3 separated the Baker Creek headwater site. **C.** K=4 separated the Big River headwater site. **D.** K=5 separated some of the Brushy Creek headwater site

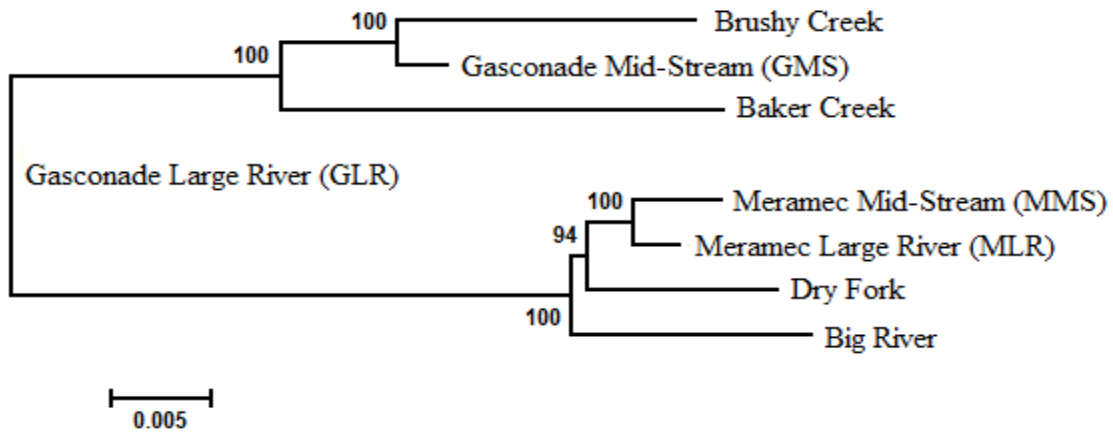
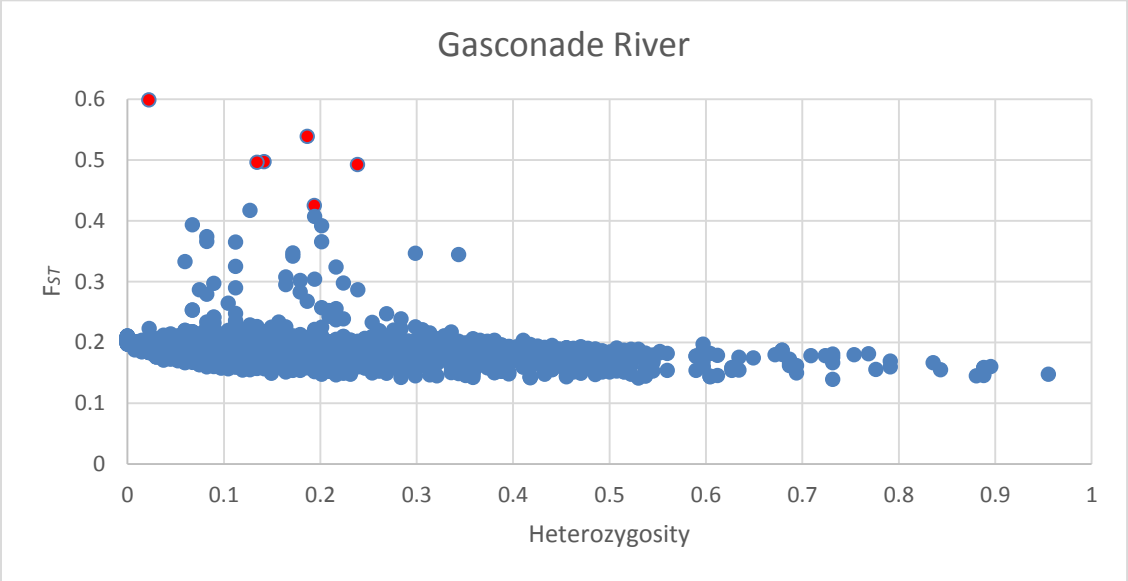


Figure 6. Neighbor Joining Phylogenetic Tree. Represents relationships between sampled sites and drainages. All bootstrap values were greater than 90% and scale represents distance of genetic change.

Outlier Loci

A genome scan for F_{ST} outliers using BayeScan, a Bayesian approach, with a prior probability of 10 produced relatively few candidate outlier loci. The goal in scanning the genome for outlier loci is to identify genomic regions under divergent selection between headwater and large river habitats. F_{ST} values were compared to levels of heterozygosity and it clearly indicates that the Gasconade drainage has moderately greater levels of divergence than the Meramec (Figure 7). In the Meramec drainage, a total of four outlier loci were identified as having a posterior odds (PO) greater than 1, which suggests a minimal signature of selection. Two loci exhibit divergence in headwater habitats, while the other two exhibit divergence in the large river habitats (Figure 8). In the Gasconade drainage, a total of six candidate outlier loci identified with posterior odds greater than 1. The headwater site in Baker Creek, MO exhibited divergence in all six outlier loci detected (Figure 8). No loci were detected as outliers in both the Meramec and Gasconade drainages.

A.



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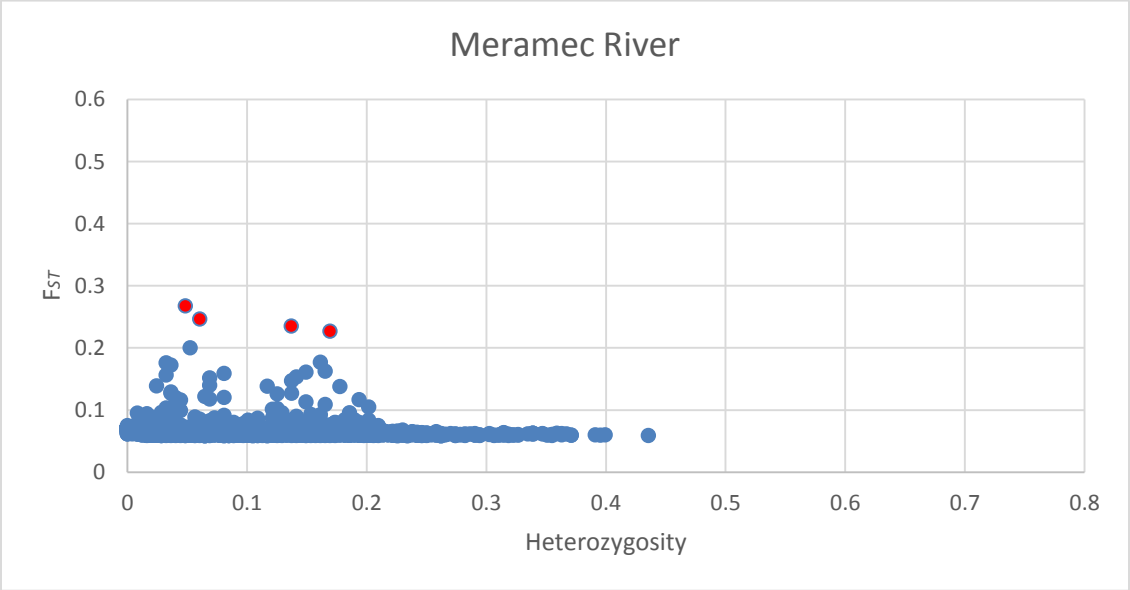
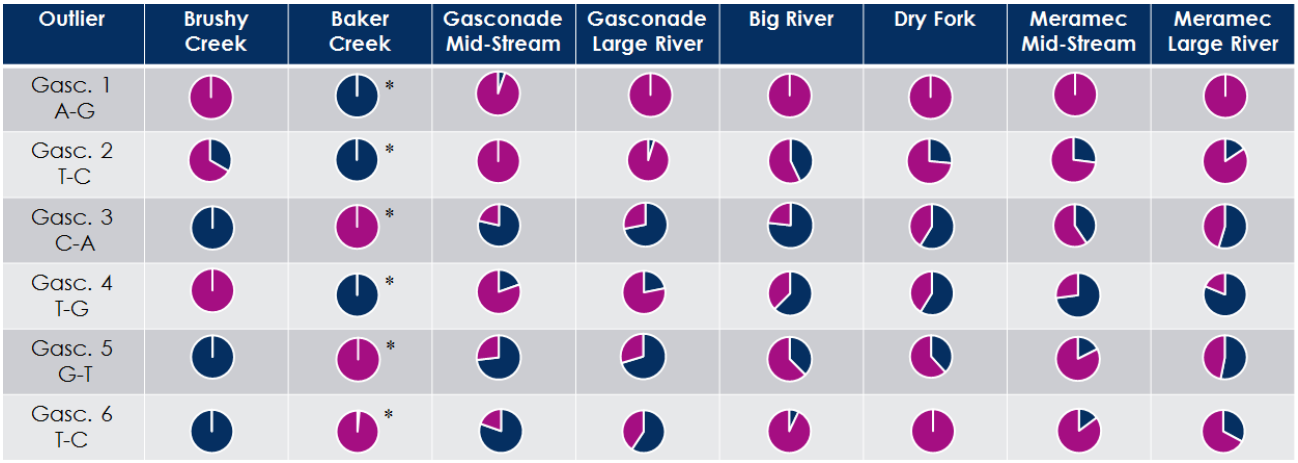


Figure 7. Plot of F_{ST} Vs Heterozygosity. **A.** Gasconade drainage. **B.** Meramec drainage. Red dots designate outlier loci detected in BayeScan.

A.



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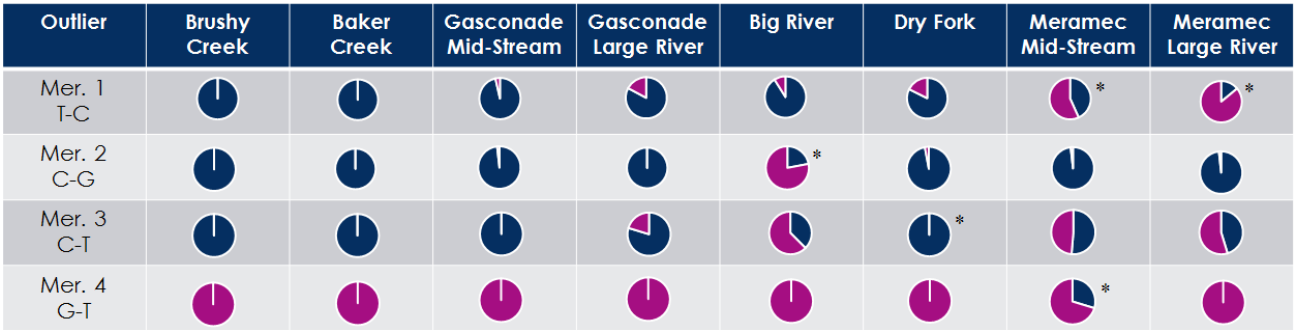


Figure 8. Polymorphic Outlier loci detected in BayeScan with corresponding allele frequencies in the **A.** Gasconade and **B.** Meramec drainages. Sites with significant divergence (*)

Geometric Morphometrics

General characteristics of morphology were observed using the geometric morphometric techniques. A sample of 100 individuals from the Gasconade drainage and 75 individuals from the Meramec drainage were used in the analysis. Young-of-year juveniles, gravid females, or fish that had excessive curvature due to preservation artifact were excluded from the study. Relative warp scores were used in constructing a PCA, then warp

scores were analyzed using three-way ANOVA to elucidate the statistical significance of each PC using the variables of sex, position within the drainage, and drainage.

Analysis

The first goal was to determine the effects of sex, position in the drainage (headwater or large river habitat), and between drainages (Gasconade and Meramec), and their interaction on body size and shape using analysis of variance (ANOVA). For both size (centroid size) and shape (PC scores), we considered sex, position, and drainage and their collective interactions.

Body Size

Overall, individuals from the Gasconade drainage were slightly larger than individuals from the Meramec drainage (Figure 9). Centroid sizes were greater in headwater habitats than large river habitats (Figure 9) and an ANOVA found centroid size to be statistically significant between the headwater and large river habitats ($p<0.001$) (Table 7). In both the Meramec and Gasconade River systems males were larger than females on average based on centroid sizes (Figure 9) but not statistically significant in the ANOVA (Table 7).

Table 7. Analysis of Variance (ANOVA) on Centroid Size. A significant effect of position within the drainage was found on centroid size.

Source of Variation on Centroid Size	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0	1.18	0.279
Position	1	0	22.34	**<0.001**
Drainage	1	0	0.44	0.509
Sex*Position	1	0	1.46	0.229
Sex*Drainage	1	0	0.44	0.508
Position*Drainage	1	0	1.88	0.172
Sex*Position*Drainage	1	0	0.61	0.435
Error	166	0		

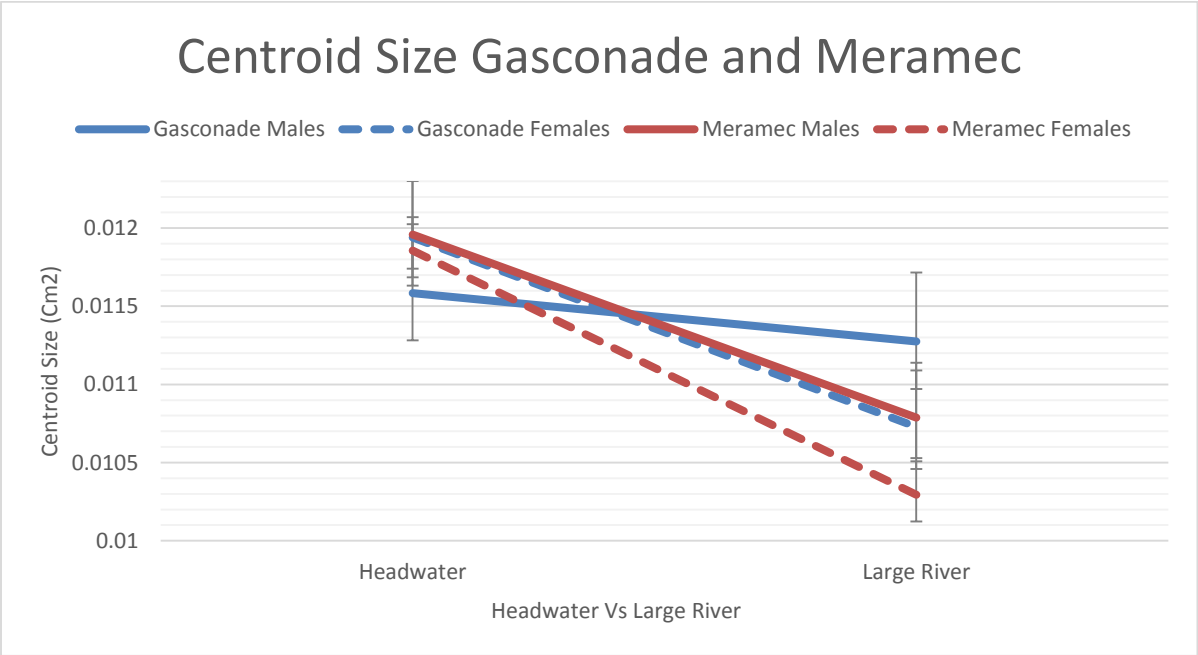


Figure 9. Bi-variate plot of males and females from the Gasconade and Meramec drainages

Body Shape

Males and females from both drainages separated along the first axis and accounted for approximately 37% of variation in overall shape. Relative warp score plots were constructed to provide visual representation of each PC. PC1 displays a general trend of males with more negative warp scores and females with more positive warp scores (Figure 10a). This variation accounted for the sexual dimorphism between the sexes and described dorsal and anal fin variation (Figure 11a & 12a). Sex ($p<0.001$) and position within the drainage ($p=0.019$) had a significant effect on PC1 (Table 8a). PC2 displays a trend with Meramec individuals with more negative warp scores and Gasconade individuals with more positive scores (Figure 10a). Body depth differences were separated along the second axis which accounted for 12% of overall shape variation (Figure 11b & 12b). Variation was found to be significant between the drainages on PC2 ($p<0.001$) (Table 8b). The effect of centroid size was found to be significant on PC2 ($p<0.001$) (Table 8b). PC3 displays a trend where Gasconade individuals have higher relative warp scores and Meramec individuals have lower relative warps scores (Figure 10b). The third axis detailed differences between caudal peduncle lengths accounting for 8% of total variation (Figure 11c & 12c). Individually, position in the drainage ($p<0.001$), and drainage ($p<0.001$) had significant effects on PC3 (Table 8c). The effect of centroid size was also found to be significant on PC3 ($p<0.001$) (Table 8b).

Table 8. Analysis of Variance (ANOVA) on Full Body **A.** Sex had a significant effect on PC1. **B.** Differences between drainages and centroid size had a significant effect on PC2 **C.** Position in the drainage, between drainages, and centroid size had an effect on PC3.

A.

Source of Variation PC1 (37%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.039	486.73	**<0.001**
Position	1	0.000	5.6	**0.019**
Drainage	1	0.000	0.86	0.354
Sex*Position	1	0.000	0.41	0.354
Sex*Drainage	1	0.000	0.81	0.523
Position*Drainage	1	0.000	2.56	0.370
Sex*Position*Drainage	1	0.000	2.62	0.107
Centroid	1	0.000	0.84	0.360
Error	165	0.000		

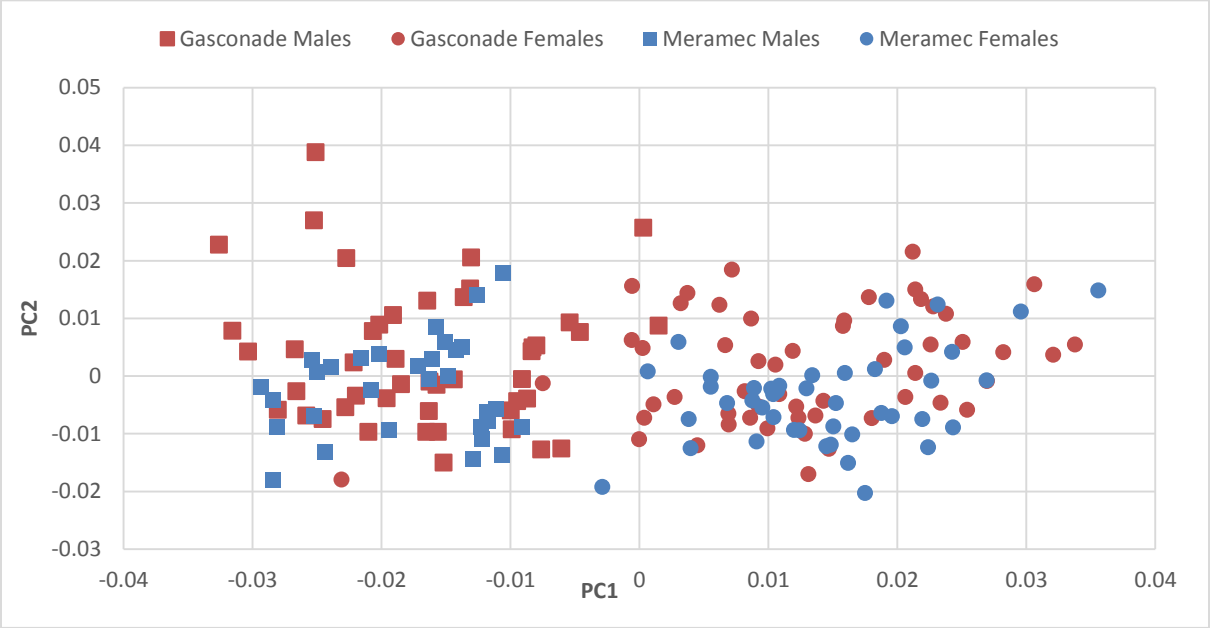
B.

Source of Variation PC2 (12%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.000	0.06	0.808
Position	1	0.000	0.16	0.691
Drainage	1	0.001	13.72	**<0.001**
Sex*Position	1	0.000	1.48	0.225
Sex*Drainage	1	0.000	0.85	0.358
Position*Drainage	1	0.000	0.64	0.426
Sex*Position*Drainage	1	0.000	0.34	0.562
Centroid	1	0.003	31.56	**<0.001**
Error	165	0.000		

C.

Source of Variation PC3 (8%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.000	3.62	0.059
Position	1	0.001	13.77	**<0.001**
Drainage	1	0.003	76.6	**<0.001**
Sex*Position	1	0.000	0.53	0.468
Sex*Drainage	1	0.000	0.27	0.602
Position*Drainage	1	0.000	1.41	0.237
Sex*Position*Drainage	1	0.000	5.09	0.025
Centroid	1	0.001	15.53	**<0.001**
Error	165	0.000		

A.



B.

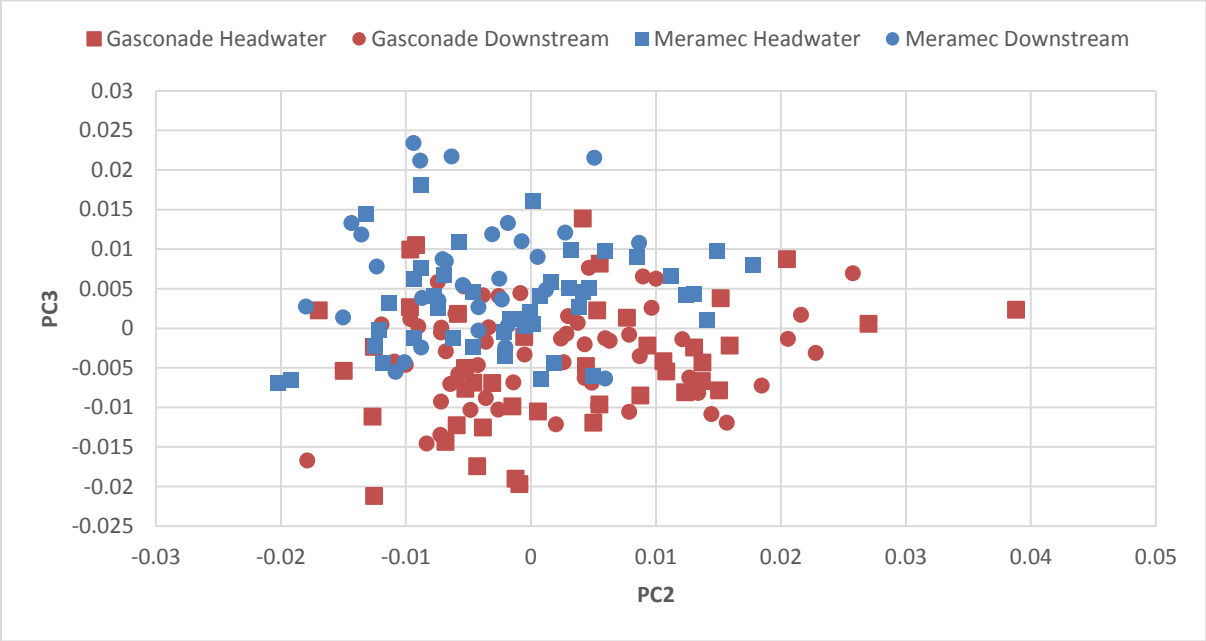
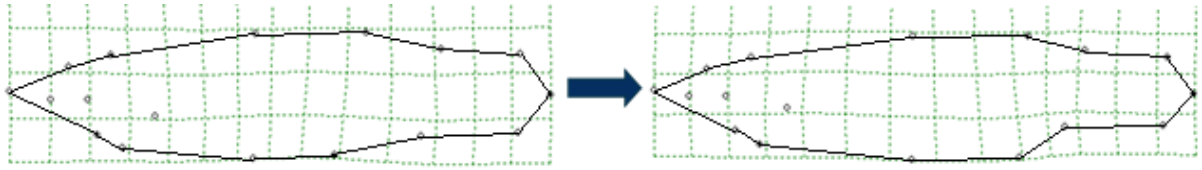


Figure 10. Principal Components Analysis (PCA) of Relative Warp Scores from Full Body. **A.** PC1 separates based on sexual dimorphism (37%). PC2 accounted for differences in deep and shallow body forms (12%). **B.** PC3 accounted for differences in caudal peduncle lengths (8%).

A.



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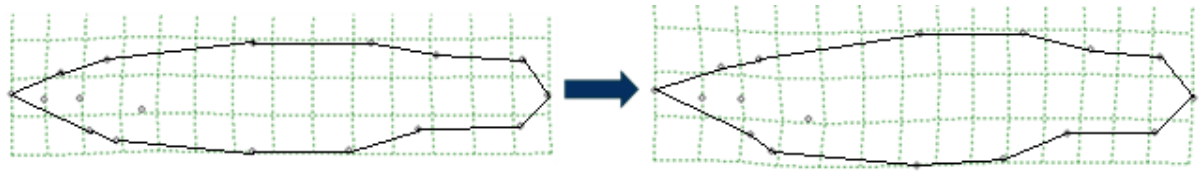


C.

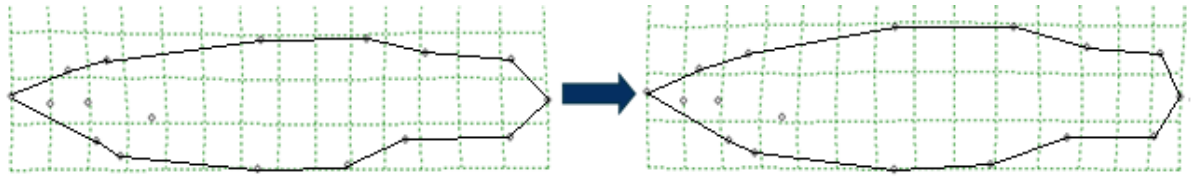


Figure 11. Thin plate spline representation of full body shape changes along **A.** PC1, **B.** PC2, **C.** PC3. Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

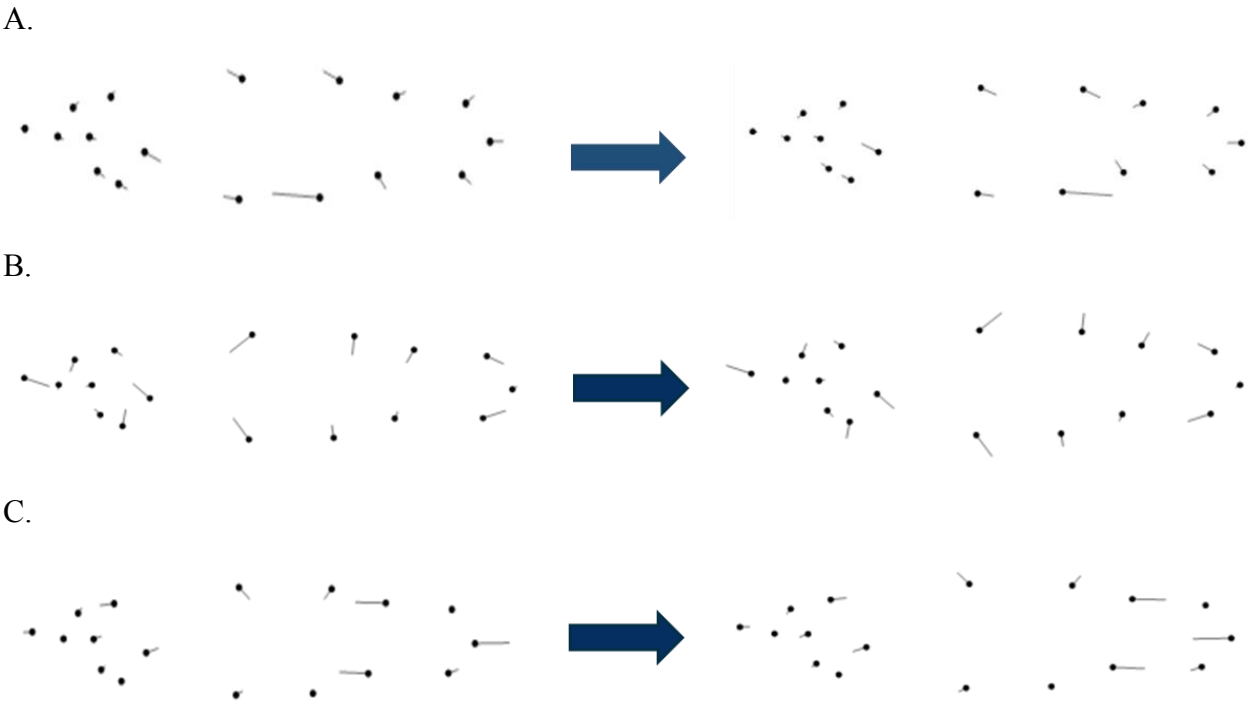


Figure 12. Shape deformation representation of full body shape changes along **A.** PC1, **B.** PC2, **C.** PC3, Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

Head, Body, and Caudal Peduncle Shape Analysis

Regions of the body were examined to assess if there were small scale differences possibly missed by the full scale analysis. In the head region, approximately 66% of the variation was described in the first two principal component axes. PC1 (38.25%) (Figure14a & 15a) described variation in the snout where headwater individuals had an elongated, more streamlined head with a downturned mouth and large river individuals had a shorter, less streamlined snout with an upturned mouth ($p<0.001$) (Table 9). PC2 (28%) described variation in location of the eyes in relation to mouth position (Figure 14b & 15b). Sex,

drainage, position in the drainage, and position by drainage were a statistically significant factors on PC2 ($p<0.01$) (Table 9).

Table 9. Analysis of Variance on Head Region. **A.** Position in the drainage is a significant variable acting upon PC1. **B.** Sex, position and, drainage act upon PC2 individually as well as position*drainage. ($p<0.05$)

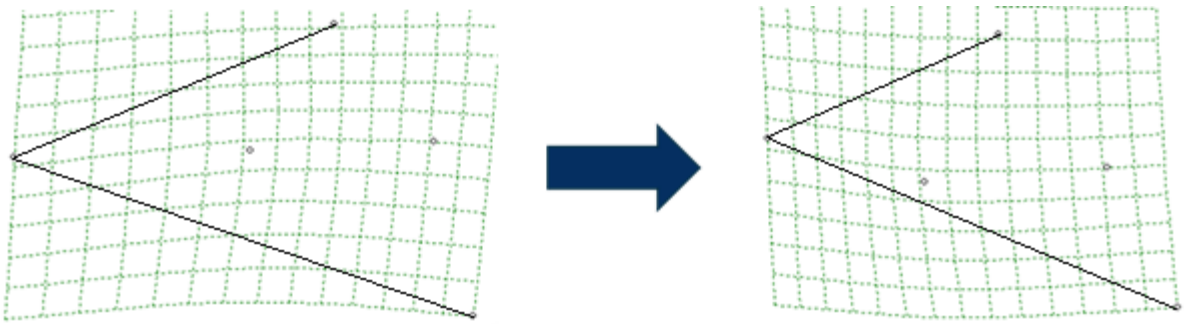
A.

Source of Variation PC1 (38%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.002	1.62	0.205
Position	1	0.018	15.34	**<0.001**
Drainage	1	0.000	0	0.957
Sex*Position	1	0.001	0.54	0.466
Sex*Drainage	1	0.000	0	0.957
Position*Drainage	1	0.000	0	0.970
Sex*Position*Drainage	1	0.000	0.07	0.794
Error	167	0.001		

B.

Source of Variation PC2 (28%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.007	11.21	**<0.001**
Position	1	0.005	7.5	**0.006**
Drainage	1	0.025	40.83	**<0.001**
Sex*Position	1	0.001	1.21	0.272
Sex*Drainage	1	0.000	0.19	0.667
Position*Drainage	1	0.014	22.54	**<0.001**
Sex*Position*Drainage	1	0.001	1.19	0.277
Error	167			

A.



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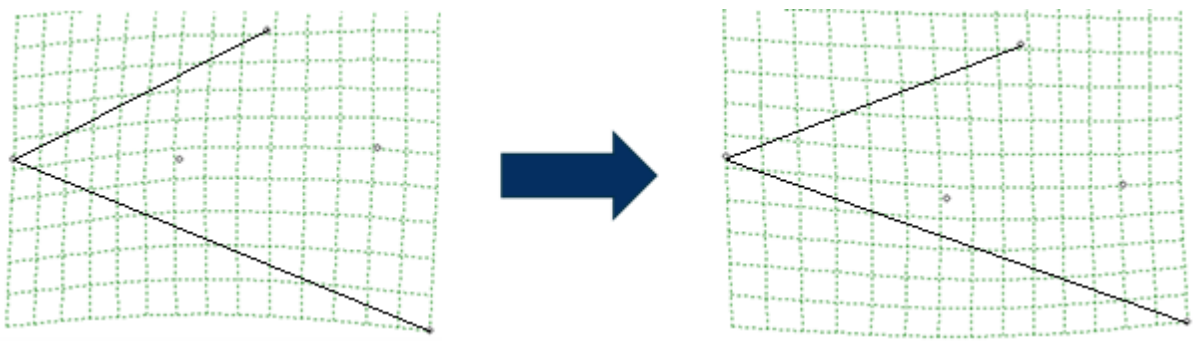


Figure 13. Thin plate spline representation of head region shape changes along **A.** PC1, **B.** PC2. Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

A.

B.

Figure 14. Shape deformation representation of head region shape changes along **A.** PC1, **B.** PC2. Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

In the body region, approximately 66% of the variation was described in the first two principal component axes. PC1 (42.5%) describes deep and shallow body forms which were accounted for in the full scale body analysis (Figure 16a & 17a). Sex ($p<0.001$) and position by drainage ($p=0.01$) significantly differed PC1 (Table 10). PC2 (23.5%) was not statistically significant for any tested variables (Table 10).

Table 10. Analysis of Variance on Body Region. **A.** Significant variables acting on PC1 include sex and position*drainage. **B.** No significant variables are acting upon PC2. ($p<0.05$)

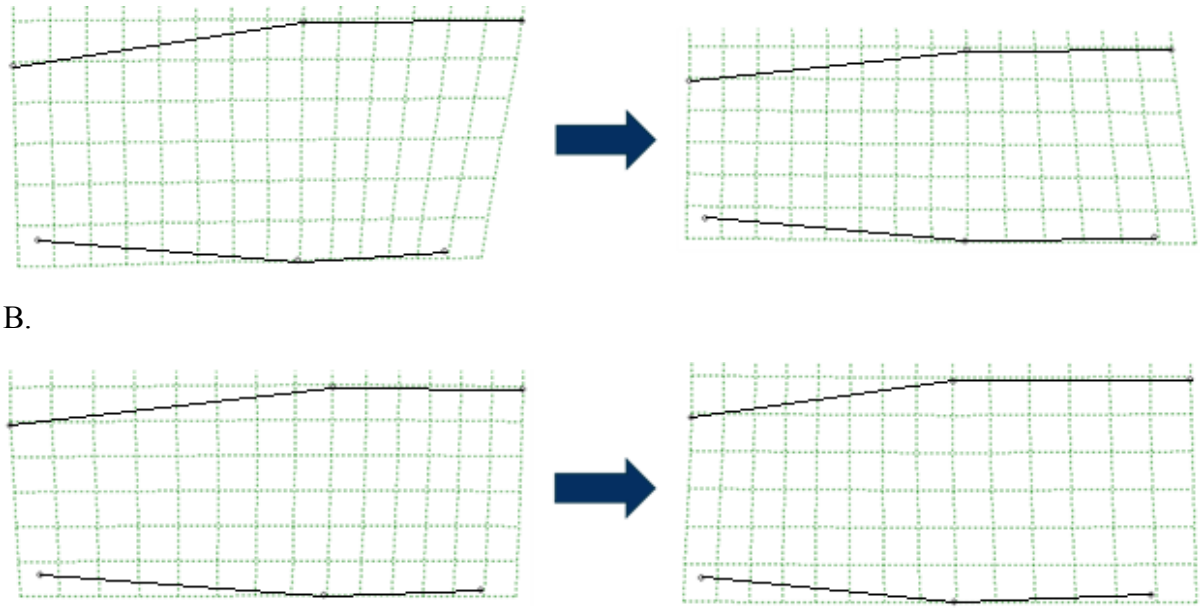
A.

Source of Variation PC1 (42.5%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.081	302.59	**<0.001**
Position	1	0.000	0.195	0.659
Drainage	1	0.000	1.25	0.265
Sex*Position	1	0.000	0.15	0.704
Sex*Drainage	1	0.000	0.53	0.468
Position*Drainage	1	0.002	6.9	**0.009**
Sex*Position*Drainage	1	0.000	0.79	0.377
Error	167	0.000		

B.

Source of Variation PC2 (23.5%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.000	0.03	0.856
Position	1	0.000	0.29	0.592
Drainage	1	0.001	2.42	0.121
Sex*Position	1	0.000	0.07	0.788
Sex*Drainage	1	0.000	0.162	0.688
Position*Drainage	1	0.000	0.01	0.958
Sex*Position*Drainage	1	0.001	2.8	0.096
Error	167	0.000		

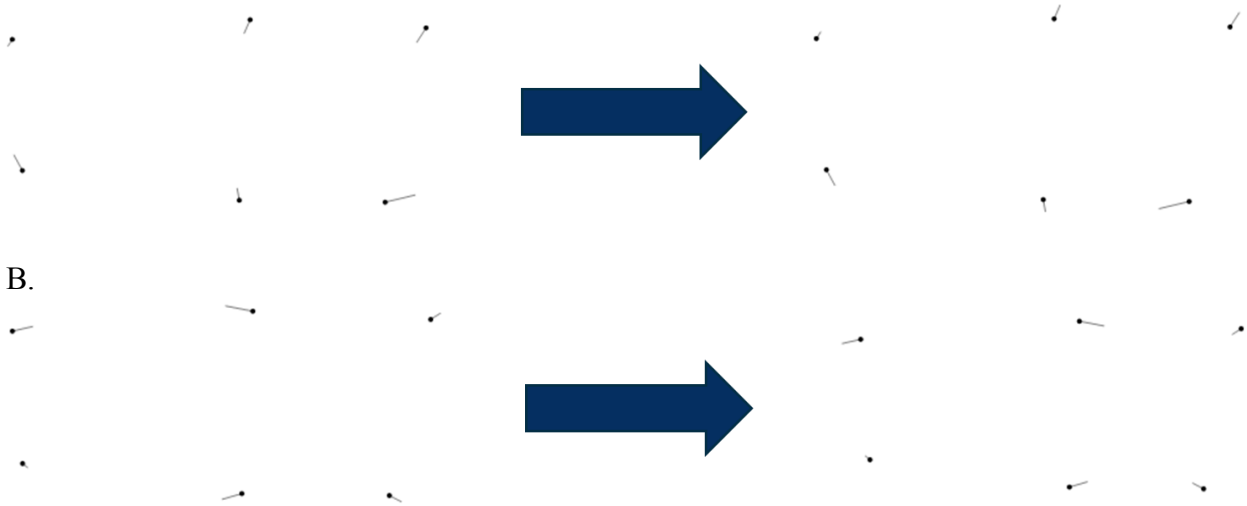
A.



B.

Figure 15. Thin plate spline representation of body region shape changes along **A.** PC1, **B.** PC2. Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

A.



B.

Figure 16. Shape deformation representation of body region shape changes along **A.** PC1, **B.** PC2. Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

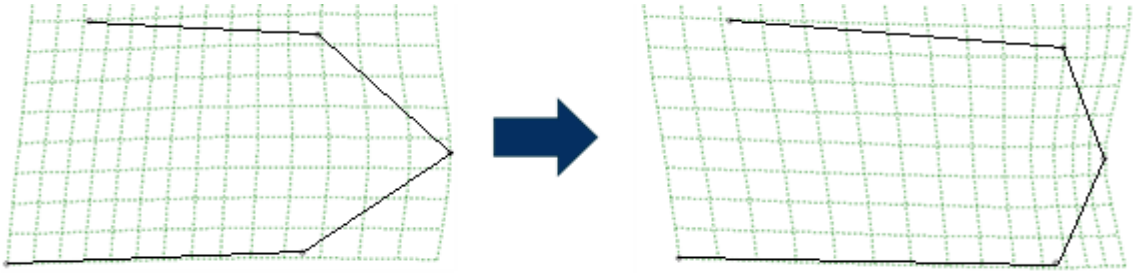
In the caudal peduncle region, approximately 64% of the variation was explained in the first two principal component axes. PC1 (35%) describes the articulation point of the tail fin (Figure 18a & 19a). Sex (p=0.002), and drainage (p<0.001) significantly differences PC1 as well as position*drainage (p=0.035) (Table 11). Males generally showed a shorter caudal peduncle than females, and the Meramec individuals had shorter caudal peduncles than the Gasconade individuals. Headwater fish within the Gasconade drainage had longer caudal peduncles, but in the Meramec drainage, large river fish had longer caudal peduncles. PC2 (29%) describes the length of the caudal peduncle region (Figure 18b & 19b). Sex (p<0.001) and drainage (p<0.001) were significant variables affecting PC2 (Table 11).

Table 11. Analysis of Variance on Caudal Peduncle Region. **A.** Significant variables acting upon PC1 include sex, drainage, and position*drainage. **B.** Significant variables affecting PC2 include sex and drainage. (p<0.05)

Source of Variation PC1 (35%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.013	10.2	**0.002**
Position	1	0.004	2.85	0.094
Drainage	1	0.019	15.27	**<0.001**
Sex*Position	1	0.000	0.03	0.863
Sex*Drainage	1	0.000	0.06	0.808
Position*Drainage	1	0.006	4.5	**0.035**
Sex*Position*Drainage	1	0.001	0.68	0.412
Error	167	0.001		

Source of Variation PC2 (29%)	Degrees of Freedom	Mean Squares	F-Ratio	p-value
Sex	1	0.050	62.62	**<0.001**
Position	1	0.002	2.61	0.108
Drainage	1	0.024	29.8	**<0.001**
Sex*Position	1	0.000	0.26	0.609
Sex*Drainage	1	0.001	1.37	0.243
Position*Drainage	1	0.000	0.49	0.485
Sex*Position*Drainage	1	0.001	1.58	0.211

A.



B.

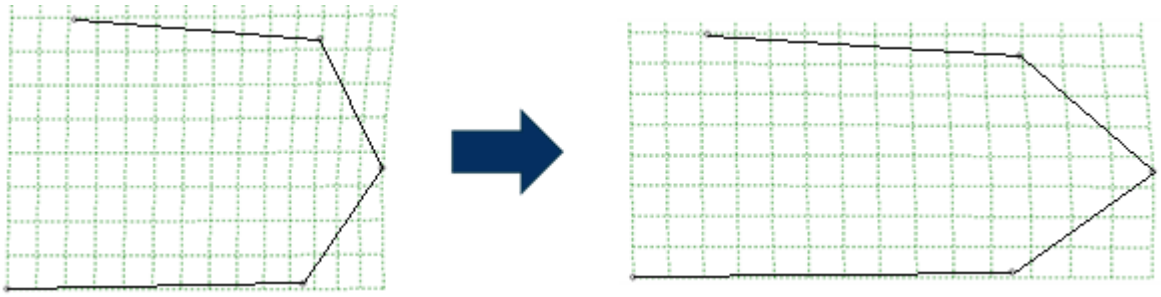
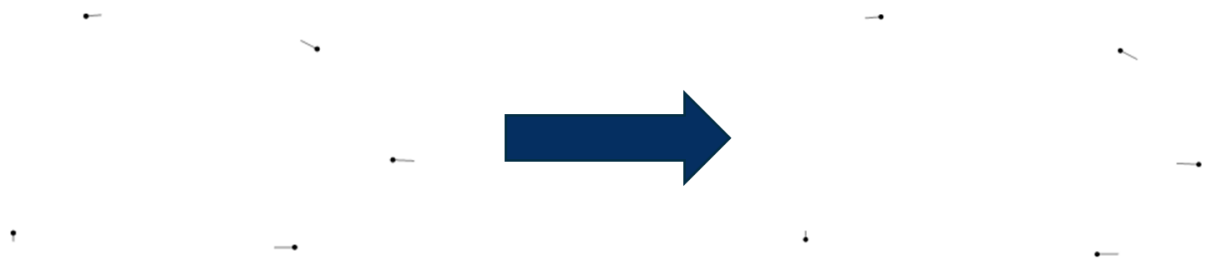


Figure 17. Thin plate spline representation of caudal region shape changes along **A.** PC1, **B.** PC2. Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

A.



B.

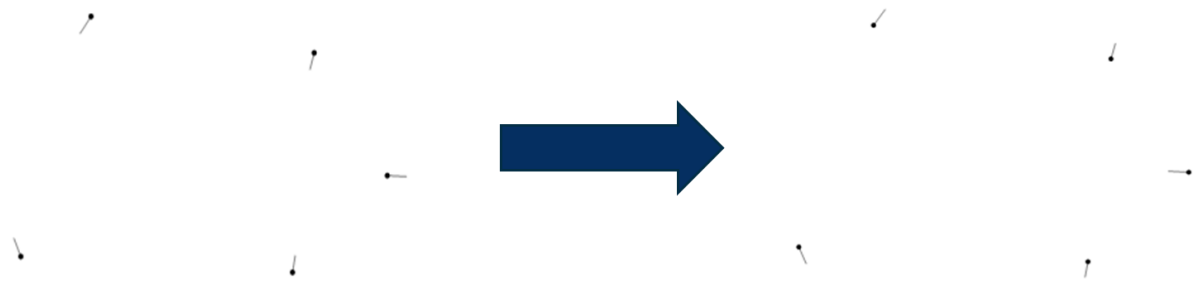


Figure 18. Shape deformation representation of the caudal region shape changes along **A.** PC1, **B.** PC2. Representations on the left side of the arrow indicate negative PC scores and those on the right side indicate positive PC scores.

CHAPTER V

DISCUSSION

Variation With and Among Drainages

The main goal of this study was to quantify the impact of neutral and non-neutral processes that partition genetic variation among populations of small stream fish and to evaluate how habitat may influence body size and shape. The Meramec and Gasconade river drainages provide a good system to study these processes due to their similarity in size, scale and physiographic regions. The topminnow is a good organism for study in this system because populations are distributed throughout the full spectrum of habitats within the river continuum from intermittent headwaters to large rivers. Our genome wide SNP data indicate general divergence between populations isolated in the two drainages with some evidence of mixed ancestry among individuals sampled in the Gasconade Large River site. One possible explanation for the mixed ancestry stems from incomplete lineage sorting. Incomplete lineage sorting could have occurred because large river environments likely contain greater effective population sizes. Recent phylogeographic studies suggest *F. olivaceus* populations in the northern portion of the species range have formed relatively recently through range expansion (Duvernell et al., 2013). A second possible explanation for mixed ancestry is due to admixture from secondary contact, possibly through an undocumented headwater capture event or via dispersal, perhaps along the Missouri River connection. A headwater capture event seems more plausible than dispersal along the Missouri River given that topminnows are not found in the Missouri River proper.

One observation that is quite interesting is that there were clear differences in levels of divergence among populations within respective drainages. Although not quantified in

this study, habitats appear to be similar between drainages in both composition and geographic distance, providing no clear explanation as to why Gasconade sites display more genetic divergence than the Meramec. The neighbor-joining phylogeny, where *Da* distance branch lengths were considerably longer and displayed much more genetic structuring among the Gasconade drainages as opposed to the Meramec drainages. The PC analysis of the genetic data across PC1 and PC2 support the results found in the neighbor-joining phylogeny. Also, F_{ST} values distributed within the Gasconade River are moderately larger than values observed in the Meramec River drainage. If the Gasconade and Meramec River drainages were colonized relatively recently (post-Pleistocene) as suggested by Duvernell et al. (2013), then the greater overall divergence observed in the Gasconade River drainage may indicate an earlier colonization than the Meramec River drainage. Geographic structuring over this longer period of time could have allowed the Gasconade drainages to establish migration-drift equilibrium. This suggests the Meramec drainage populations are not yet in equilibrium with respect to gene flow and genetic drift. Recent range expansion could have played a significant role in genome-wide patterns of genetic diversity where founder effects, bottlenecks, and genetic drift can be especially strong during expansions (Excoffier et al., 2009; Ramachandran et al., 2005).

We predicted that headwater populations in each site would either diverge independently of one another through neutral evolutionary processes or all headwater populations would be impacted by non-neutral processes in the same manner and exhibit parallel divergence from adjacent large river populations. Our analyses indicated a distinct genetic signature in three of four (Brushy Creek, Baker Creek, Big River) headwater populations. The individual divergence seen in each of these small tributary streams supports

our hypothesis that neutral processes are impacting each headwater population independently and diverging from adjacent large river populations, as well as other headwater populations. In addition to headwater divergence, we expected headwater populations to exhibit reduced genetic diversity due to smaller population sizes, propensity to undergo periodic bottleneck events, and geographic isolation from large river populations due to the dendritic nature of river systems. Such a pattern has been seen in studies of other stream species. For example, a study by Junker et al. (2012), demonstrated a significant loss in genetic diversity in headwater European bullhead (*Cottus gobio*) populations stemming from geographic isolation. Our analyses indicated that if headwater populations have been subjected to more bottleneck events, such events have not substantially impacted genetic diversity. Measures of genetic diversity in headwater sites were only slightly lower than those observed in large river sites.

Outlier Loci

The second objective to the genome-wide analysis was to determine if headwater sites are under different selective pressures than large river sites and if there are genomic signatures of selection. One method in determining selection is to identify outlier loci, which are loci in which allele frequencies significantly deviate from neutral, non-outlier background levels. Our genome-wide analysis identified only a small number of presumptive SNP outlier loci. In fact, this study uncovered fewer candidate outlier loci than is typically observed in other similar studies (Deagle et al., 2012; Pujolar et al., 2014; Seeb et al., 2014). Outlier tests can often be a weak method of identifying outlier loci due to the use of F_{ST} based genome scans and simulation studies have shown a propensity for identifying false positives (Villemereuil et al., 2014). Given the small numbers of outliers observed, our data

do not provide any indication of divergent selection pressures between headwaters and large river habitats on genomic regions sampled.

Morphological Variation in Size and Shape

One goal of the morphological study was to explore body size differences of *F. olivaceus* between headwater and large river populations. In both the Gasconade and Meramec River drainages we found evidence in support of our hypothesis that body size was associated with habitat, where headwater populations had larger overall size than large river populations. There are a few explanations that could account for the size differences observed between headwater and large river populations. First, predation is believed to be the driving pressure selecting for smaller size at maturation in large river habitats (Charlesworth, 1980; Conover & Munch, 2002). We attempted to control for seasonal conditional effects by collecting all samples within a two week period, but due to unseasonable flooding in the Meramec large river site near Arnold, Missouri, we had to sample a month later. The significant differences in size between headwater and large river sites could also be attributed to life history differences. For example, Reznick & Bryga (1996) demonstrated that guppies in high predation sites mature at an earlier age and smaller size, shifting energetic efforts to reproduction rather than growth. Secondly, the age structure may differ between sites such that headwater fish are older on average as a results of reduced mortality from predatory piscivorous fish (Larimore & Bailey, 1996; Langerhans et al., 2004). Additionally, increased nutrient inputs in headwater environments may allow individuals in these environments to grow larger than large river individuals (Vannote et al., 1980). Lastly, water is generally clear in headwater environments potentially increasing the ability for females to choose larger mates. Female choice has been documented in guppies,

where females select for larger males and enhance reproductive fitness for larger individuals (Bischoff et al., 1985). We cannot make a clear inference whether body size is driven by local adaptation, life history characteristics, age, nutrient inputs, or simply a function of phenotypic plasticity. Future studies using simulated environments for common garden experiments may be able to address this issue.

The second goal of the morphological study was to determine body shape variation between headwater and large river individuals. We expect characteristics of headwater fish to be more streamlined for swimming in fast flowing waters and large river fish to be adapted for better maneuverability. Based upon expectations from Schaefer et al. (2011), the sister species, *F. notatus*, generally displayed a deep body form when found in headwater habitats and shallow body forms in large river habitats. Deep body forms allow the fish to be more streamlined for easier sustained swimming in faster flow regimes (Webb, 1984). The Schaefer et al. (2011) study demonstrated that *F. olivaceus* did not take on a shallow body form as *F. notatus* did in downstream large river environments. We predicted that when *F. olivaceus* solely occupied the river system from headwaters to large rivers, *F. olivaceus* would converge on a similar body shape to *F. notatus*, displaying deep body forms in headwater environments and shallow body forms in large river environments. Our findings did not support our hypothesis and indicates that *F. olivaceus* does not take on a significantly shallower body form in large river environments. Body depth was found to be significantly different between the Meramec and Gasconade drainages, but not among habitat types within drainages. It appears that *F. olivaceus* shows less geographic variation for this characteristic than *F. notatus* and vary to a lesser degree.

In the head region, we expected individuals in headwater habitats to exhibit a more streamlined head shape for optimal sustained swimming and a less streamlined head designed for maneuverability in large river habitats. We did find evidence supporting our hypothesis, in which headwater fish exhibited longer, more streamlined snouts, where large river fish exhibited more inferior snouts. Thomerson (1966) analyzed a series of meristic and morphological traits in *F. notatus* and *F. olivaceus* and found that some measures such as head shape, snout length, and body depth were different between the species in a given river, but also noted flow regime differed by river and was variable across the range. Our data agrees with Thomerson's account that flow regime in differing drainages may account for noticeable changes in morphological characteristics, such as head shape and snout length.

Sexual dimorphic characteristics are quite common among many species and significantly impacts body shape, so sexually dimorphic traits were expected to be observed in the analysis (Parker, 1992; Adams et al., 1997; Kaliontzopoulou et al., 2007). As expected, sex substantially influenced overall body shape in two of four principal components, suggesting a high degree of sexual dimorphism in topminnows. Welch et al. (2013) described male topminnows having larger traits, such as anal and dorsal fin base lengths. The anal and dorsal fins of topminnows are important structures used for sex-specific roles in courtship and spawning (Carranza & Winn, 1954). Sex had a significant influence on caudal peduncle shapes as well, where males had shorter caudle peduncles than females. The length of the caudal peduncle might be explained by the length of the anal fins of males, causing the caudal peduncle region to appear shorter. Centroid size was used as a covariate in the analysis of variance, and the caudal peduncle length seemed to have an

allometric relationship with overall size. Larger fish exhibited greater characteristics, where smaller fish exhibit lesser characteristics.

Overall, subtle shape differences among populations of topminnows were observed between drainages, and headwater and large river environments. Aside from sexually dimorphic characteristics, the variances observed are consistent with differences in flow regime, where headwater topminnows display characteristics favoring a streamlined body (streamlined snout, caudal peduncle length) allowing them to efficiently swim in fast flowing waters. Headwater fish also exhibited a greater overall size than large river fish which may be attributed to the age of the fish or habitat specific factors, such as predation, or maturation at smaller sizes.

Additional common garden experimentation by rearing young in differing flow regimes may provide insight into whether the trait had varying degrees of plasticity or if there is a genetic component. Higher resolution genomic sampling could potentially reveal loci under selection identifying genes that are known to affect body size and shape to better understand selective pressures in the river continuum.

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